MUSCLE OS

PILLAR 1

Diet & Nutrition

the complete guide to body composition

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Domain 1: Nutritional Foundations

1. Energy Balance & Energy Availability

1.1 The First Law Applied to Body Composition

Energy balance is the application of the first law of thermodynamics to living systems: changes in body energy stores equal energy intake minus energy expenditure. This forms the non-negotiable foundation of any body composition intervention. No nutritional strategy can bypass this fundamental equation; it merely shifts the composition of what is gained or lost.

Key Distinction

Energy balance determines **whether** weight changes. The P-Ratio determines **what** that weight change is made of (fat vs. lean mass). Both must be managed simultaneously.

1.2 Total Daily Energy Expenditure

TDEE is the sum of four components:

Table 1.1 Components of Total Daily Energy Expenditure

COMPONENT	DESCRIPTION	TYPICAL % OF TDEE
Basal Metabolic Rate (BMR)	Energy required for basic physiological functions at rest	60–75%

Thermic Effect of Food (TEF)	Energy cost of digestion, absorption, and nutrient storage	8–12%
Non-Exercise Activity Thermogenesis (NEAT)	Energy from all non-exercise movement (fidgeting, walking, posture)	15–30%
Exercise Activity Thermogenesis (EAT)	Energy from deliberate exercise	5–15%

1.3 Energy Availability vs. Energy Balance

Energy availability (EA) separates physiological health from thermodynamics. EA is defined as dietary energy remaining for physiological functions after subtracting the energy cost of exercise:

$$\text{EA} = \frac{\text{Energy Intake (kcal)} - \text{Exercise Energy Expenditure (kcal)}}{\text{kg Fat-Free Mass}}$$

Table 1.2 Energy Availability Zones

ZONE	VALUE (KCAL/KG FFM/DAY)	PHYSIOLOGICAL IMPACT
Optimal	>45	Full physiological function, optimal hormonal health
Reduced	30–45	Possible menstrual disruption, metabolic adaptation begins
Low	<30	Physiological suppression, hormonal dysfunction, LEA syndrome

1.4 Metabolic Adaptation

Prolonged caloric restriction triggers metabolic compensation: reduced NEAT, reduced TEF, reduced activity in the sympathetic nervous system, and hormonal downregulation (leptin, T3, testosterone). This is not metabolic damage—it is adaptive thermogenesis. Expect a 10–15% reduction in TDEE beyond what is predicted by weight loss alone. Reversal requires a structured reverse-dieting approach (typically 50–100 kcal increases per week over 4–12 weeks).

CHAPTER SUMMARY

Energy balance determines whether weight changes; the P-Ratio determines what that change is made of. A coach must understand the four TDEE components (BMR, TEF, NEAT, EAT) and distinguish energy balance from energy availability. Metabolic adaptation is a compensatory response, not damage — expect a 10–15% TDEE reduction beyond weight-loss predictions. Before adjusting calories, audit tracking accuracy and confirm the deficit is truly being followed.

◈ MAKE THIS WORK FOR YOU

Your TDEE is not a fixed number — it shifts with your activity, stress, sleep, and dieting history. A desk-bound professional maintaining 2,400 kcal may drop to 2,100 after 8 weeks of dieting, not because they stopped trying, but because NEAT drifted downward and metabolic adaptation kicked in. If you have been dieting for more than 12 weeks or lost more than 10% of your starting body weight, expect your TDEE to be 10–15% lower than what equations predict.

For the beginner: Use the Mifflin-St Jeor equation as a starting point but treat it as a hypothesis, not a verdict. Track your weight for 2–3 weeks at your estimated maintenance, then adjust based on the actual trend.

For the experienced dieter: Do not chase an ever-larger deficit when progress stalls. Audit tracking accuracy first, then check NEAT (steps have likely dropped), then consider a diet break at maintenance for 2–4 weeks to restore leptin and thyroid function before resuming the deficit.

2. The P-Ratio & Nutrient Partitioning

2.1 Defining the P-Ratio

The P-Ratio (Protein-to-Fat partitioning ratio) describes the proportion of an energy imbalance that is deposited as or mobilised from lean mass versus fat mass. In a surplus, it determines how much weight gain is muscle versus fat. In a deficit, it determines how much weight loss is fat versus muscle. Forbes (1987) established the canonical equation describing the relationship between body fat percentage and the P-Ratio.

2.2 The Five Determinants of the P-Ratio

Table 2.1 Determinants of Nutrient Partitioning

DETERMINANT	MECHANISM	EFFECT ON P-RATIO
Protein intake	Leucine signalling via mTOR, substrate provision	Higher protein improves lean mass retention/gain
Training stimulus	MPS elevation, GLUT4 translocation, insulin sensitivity	Strong training signal partitions nutrients toward muscle
Sleep & recovery	Cortisol regulation, GH pulsatility, glycogen resynthesis	Poor sleep worsens partitioning
Metabolic adaptation	Reduced NEAT, T3 suppression, leptin decline	Chronic deficit worsens partitioning
Allostatic load	HPA axis activation, cortisol elevation	High stress impairs lean mass retention

2.3 Energy Flux: The Partitioning Advantage

Energy flux refers to the rate at which energy flows through the system (high intake + high expenditure vs. low intake + low expenditure at the same energy balance). Higher flux states (e.g., eating more and exercising more to maintain weight) produce superior partitioning outcomes compared to lower flux states at the same energy balance. This is mediated by improved insulin sensitivity, higher MPS rates, and greater substrate cycling.

Practical Application

For a client maintaining at 2,500 kcal, prefer 3,000 kcal intake + 500 kcal exercise over 2,000 kcal intake + 0 kcal exercise. Both produce the same energy balance, but the higher-flux state produces better body composition outcomes.

2.4 Nutrient Sublimation

When energy surplus exceeds the storage capacity of muscle glycogen, intramuscular triglyceride, and adipose tissue, the body activates disposal pathways: diet-induced thermogenesis (DIT), de novo lipogenesis (DNL), and substrate cycling (futile cycles). This "metabolic overflow valve" limits net energy storage but at a metabolic cost. The effectiveness of sublimation varies by individual and is influenced by insulin sensitivity, training status, and genetic factors.

CHAPTER SUMMARY

The P-Ratio determines whether a calorie surplus builds muscle or fat, and whether a deficit spares muscle while losing fat. The five key determinants are protein intake, training stimulus, sleep, metabolic adaptation, and allostatic load. Higher energy flux (high intake plus high expenditure) produces superior partitioning at the same energy balance. Coach action: prioritise protein, training, and sleep before manipulating calories.

◈ MAKE THIS WORK FOR YOU

Your personal P-Ratio depends heavily on where you are starting from. A lean individual at 10% body fat will partition a calorie surplus toward muscle more efficiently than someone at 25% body fat — the Forbes curve is not abstract, it is a description of your biology. If you are in a deficit and losing strength, your P-Ratio has shifted unfavourably, and protein, training stimulus, or sleep need attention.

If you are in a surplus: Keep the rate of gain slow (0.5–1% of body weight per month) to minimise fat accumulation. The surplus does not need to be aggressive — a modest 200–300 kcal/day above maintenance, combined with high protein and consistent training, will drive muscle gain with minimal fat.

If you are in a deficit: High energy flux is your best friend. Rather than dropping calories very low, add a modest amount of cardio (2,000–3,000 extra steps or 15–20 minutes of incline walking) so you can eat more while maintaining the same deficit. This keeps training performance higher and partitioning better.

If you are older (45+): Anabolic resistance means your P-Ratio will naturally be less favourable. Counter this with higher protein per meal (30–40 g), consistent resistance training, and prioritising sleep quality — these modifiable factors become more important with age.

3. Protein Sufficiency & the Leucine Threshold

3.1 The Molecular Cascade

Dietary protein stimulates muscle protein synthesis (MPS) through a well-characterised molecular cascade. Leucine, the key branched-chain amino acid, activates mTORC1 via the Rag GTPase pathway. This requires a minimum leucine concentration per meal—the leucine threshold—below which no meaningful MPS stimulation occurs, regardless of total daily protein intake.

3.2 The Leucine Threshold

Table 3.1 Leucine Thresholds by Population

POPULATION	LEUCINE PER MEAL	EQUIVALENT PROTEIN PER MEAL
Young adults (18-35)	2-3 g	20-25 g high-quality protein
Middle-aged (35-60)	2.5-3 g	25-30 g high-quality protein
Older adults (60+)	3-4 g	30-40 g high-quality protein

3.3 Key Evidence

Morton et al. (2018), in a meta-analysis of 49 RCTs (n = 1,863), demonstrated that protein intake for lean mass gains plateaus at approximately 1.6 g/kg/day (95% CI extends to 2.2 g/kg/day). This is the foundational evidence for protein target setting. Pasiakos et al. (2015) showed that during caloric deficit, protein requirements increase: ≥2.0 g/kg/day (and up to 2.4 g/kg/day) is recommended for fat loss phases to preserve fat-free mass.

Morton et al. (2016) demonstrated that even distribution of protein across 4–5 meals at approximately 0.4 g/kg/meal produces superior MPS compared to skewed distribution. Aragon & Schoenfeld (2013) established that the post-exercise "anabolic window" is measured in hours, not minutes; total daily intake dominates over precise timing.

CHAPTER SUMMARY

Each meal must reach the leucine threshold (2–4 g leucine depending on age) to stimulate muscle protein synthesis. Total daily protein plateaus at 1.6 g/kg/day for gains, rising to 2.0–2.4 g/kg/day in a deficit. Even distribution across 4–5 meals at 0.4 g/kg/meal outperforms skewed intake. Coach action: ensure every meal contains at least 20–40 g of high-quality protein, and prioritise total daily intake over precise timing.

◈ MAKE THIS WORK FOR YOU

Your protein needs are not generic — they scale with your goal, age, and diet type. A vegetarian who relies on lentils and tofu needs 15–25% more total protein than an omnivore to compensate for lower leucine density. An athlete in a caloric deficit may need up to 2.4 g/kg to preserve lean mass. A client over 60 needs 30–40 g of protein per meal due to anabolic resistance, not 20 g.

If you struggle to eat enough protein: Start each meal with the protein source (eat the chicken or tofu first, then the rice and vegetables). This simple behavioural cue ensures the most important nutrient is consumed before you feel full. Batch-cook protein portions for 3–4 days at a time to reduce daily decision fatigue.

If you eat 2 meals a day (intermittent fasting or time constraints): You can still meet your protein targets, but each meal must be larger — aim for 0.5–0.6 g/kg per meal. A 70 kg person doing two meals would need approximately 50–60 g of protein per meal, which requires intentional meal design (e.g., a 200 g chicken breast or a substantial tofu stir-fry).

Pre-sleep protein: If you train in the evening or tend to go 10–12 hours overnight without eating, 30–50 g of casein (cottage cheese, Greek yogurt, or a casein shake) before bed elevates MPS throughout the night and improves overnight whole-body protein balance. This is especially valuable during a caloric deficit or for older adults.

4. Carbohydrate & Glycogen Regulation

4.1 The Two-Compartment Glycogen System

The body stores glycogen in two compartments. Muscle glycogen (300–500 g total) serves as the primary fuel for high-intensity contractions and is compartmentalised into distinct pools associated with different subcellular regions. Liver glycogen (approximately 100–120 g) maintains blood glucose primarily for central nervous system function. These compartments are regulated independently and have different refuelling dynamics.

4.2 GLUT4 Translocation: The Insulin-Independent Pathway

GLUT4 is the insulin-responsive glucose transporter that mediates glucose entry into muscle cells. Critically, there are two pathways for GLUT4 translocation:

1. **Insulin-dependent pathway** (IRS1/PI3K/Akt/AS160) — responds to carbohydrate feeding and insulin release
2. **Contraction-dependent pathway** (AMPK, CaMKII, NOX) — responds to muscle contraction during exercise, independently of insulin, creating a 24–48 hour post-training window of enhanced glucose disposal

4.3 The AMPK-mTOR Seesaw

The AMPK-mTOR relationship represents the master catabolic-anabolic kinase antagonism. AMPK activates in low-energy states (exercise, caloric deficit), suppressing mTORC1 via TSC2 and Raptor phosphorylation to conserve resources. Conversely, mTORC1 activation (feeding, growth signalling) suppresses AMPK. This creates a temporal separation that athletes exploit through nutrient timing: training in the AMPK-dominant state (fasted or low-carb), then feeding to shift to the mTOR-dominant state for recovery.

4.4 Carbohydrate Periodisation

Strategic carbohydrate timing exploits three principles: (1) training-day loading around workout windows to maximise performance and glycogen refuelling, (2) rest-day reduction to improve insulin sensitivity and fat oxidation, and (3) targeted refeeds during extended deficit phases to maintain leptin, thyroid function, and training performance. Glycogen compartments respond to depletion patterns: muscle glycogen is replenished by local substrate availability; liver glycogen is prioritised by systemic glucose demand.

CHAPTER SUMMARY

Carbohydrates fuel high-intensity training via muscle and liver glycogen, regulated independently with different refuelling dynamics. GLUT4 translocation occurs through both insulin-dependent and contraction-dependent pathways, creating a 24–48 hour post-training window of enhanced glucose disposal. Strategic periodisation — loading around training, reducing on rest days — optimises performance and insulin sensitivity. Coach action: align carbohydrate intake with training demand, not arbitrary meal timing.

◈ MAKE THIS WORK FOR YOU

Carbohydrate needs are the most individual variable in nutrition because they depend on training volume, muscle mass, insulin sensitivity, and daily activity. A powerlifter doing 3 heavy sessions per week may function well on 3 g/kg/day, while a physique athlete training 6 days per week with high volume may need 5 g/kg/day or more to sustain performance and recovery.

If you have poor insulin sensitivity (PCOS, family history of diabetes, or you store fat predominantly around the midsection): Prioritise carbohydrates around your training window and reduce them on rest days. Favour lower-glycaemic sources (oats, sweet potato, legumes) at non-training meals, and reserve higher-glycaemic options (white rice, potatoes, fruit) for post-training when GLUT4 translocation is at its peak and insulin sensitivity is naturally elevated.

If you are an endurance athlete or have very high training volume: Your carbohydrate requirements are higher and more consistent across the week. Undereating carbohydrates compromises performance more than any other macronutrient because glycogen depletion directly impairs high-intensity output. Do not follow generic low-carb advice — your training demands the fuel.

If you are on a fat loss phase: Do not drop carbohydrates to zero. Instead, periodise them: maintain moderate carbs on training days (especially before and after your workout) to preserve training performance, and reduce them on rest days. This preserves performance where it matters while improving insulin sensitivity on days when it is most beneficial.

5. Dietary Fat & Hormonal Health

5.1 Essential Fatty Acids

Dietary fat is not merely a calorie source; it provides essential fatty acids (linoleic acid, alpha-linolenic acid) that cannot be synthesised endogenously and serves as the structural precursor for all steroid hormones, including testosterone, oestrogen, and cortisol. A minimum of 0.5–0.8 g/kg/day of dietary fat is required to maintain hormonal homeostasis. Below this threshold, testosterone suppression and menstrual disruption are predictable outcomes.

5.2 Fat Types and Metabolic Effects

Table 5.1 Dietary Fat Classifications

TYPE	SOURCES	METABOLIC ROLE
Saturated fat	Animal fats, coconut oil, dairy	Precursor to steroid hormones; limit to <10% of total calories
Monounsaturated fat (MUFA)	Olive oil, avocado, nuts	Improves lipid profile; anti-inflammatory
Polyunsaturated fat (PUFA)	Fish oil, flaxseed, walnuts	Essential: source of omega-3 and omega-6 fatty acids
Trans fat	Industrial hydrogenation	Pro-inflammatory; no known benefit; minimise entirely

5.3 Omega-3 to Omega-6 Ratio

Modern diets are characterised by an excessive omega-6 to omega-3 ratio (15:1 to 20:1, versus the evolutionary ratio of approximately 1:1). Elevated omega-6 intake promotes a pro-inflammatory eicosanoid profile through arachidonic acid cascade. Correcting this balance through increased EPA/DHA intake (from fatty fish or supplementation) improves membrane fluidity, resolvin synthesis, and muscle protein synthesis responses to feeding.

CHAPTER SUMMARY

Dietary fat provides essential fatty acids and is the structural precursor for steroid hormones including testosterone. A minimum of 0.5–0.8 g/kg/day is required to maintain hormonal homeostasis. The modern omega-6 to omega-3 imbalance (15:1–20:1 versus the evolutionary 1:1) promotes inflammation and should be corrected through increased EPA/DHA intake. Coach action: never drop fat below 0.5 g/kg/day, prioritise omega-3 sources, and educate clients on fat types rather than fearing dietary fat.

◈ MAKE THIS WORK FOR YOU

Your individual fat minimum depends on your sex, activity level, and hormonal health. A female athlete or a client with a history of menstrual disruption should never drop below 0.6–0.8 g/kg/day because the hormonal cascade that supports oestrogen production (and with it, bone density, libido, and mood stability) depends on adequate dietary fat.

If you are a female athlete: Your fat minimum is higher than a male's at the same body weight. Do not let your fat intake drop below 0.6 g/kg/day, even during an aggressive fat loss phase. The trade-off between faster fat loss and hormonal disruption is not worth it — irregular cycles, reduced bone density, and mood disturbances are predictable consequences of chronically low fat intake.

If your cholesterol markers are a concern: The type of fat matters more than the amount. Replace saturated fat sources (butter, fatty cuts of red meat, palm oil) with unsaturated options (olive oil, avocado, nuts, seeds) without reducing total fat. This preserves the structural role of fat in hormone production while improving your lipid profile.

If you follow a low-carb or ketogenic diet: Your fat intake will naturally be higher (1.5–2.5 g/kg/day) because fat fills the calories that would otherwise come from carbohydrates. In this context, prioritise fat quality — emphasise MUFA and PUFA sources over saturated fat, and ensure your EPA/DHA intake is adequate (consider an algae or fish oil supplement).

COACHING APPLICATION

Domain 1 Case Study: Breaking Through a Fat Loss Plateau

Scenario: A 32-year-old male client, 85 kg, 15% body fat, has been training for 2 years. He wants to improve body composition but is stuck eating 2,200 kcal/day and not losing weight. He trains 4x/week.

Assessment: Calculate his TDEE using the Mifflin-St Jeor equation. His estimated BMR is ~1,800 kcal. With moderate activity (1.55), TDEE ≈ 2,790 kcal. He is eating well below this yet not losing weight — suggesting metabolic adaptation from chronic undereating or inaccurate tracking.

Intervention: (1) Audit food tracking accuracy — hidden oils, condiments, and portion size drift are common. (2) Increase to 2,400 kcal for 2 weeks as a "metabolic reset" while maintaining protein at 2.0 g/kg (170 g/day). (3) Add 8,000 steps/day NEAT target. (4) Reassess after 2 weeks: weight trend, energy levels, training performance.

Key Takeaway: Energy balance is foundational, but metabolic adaptation and tracking accuracy must be assessed before assuming a deficit is insufficient. The P-Ratio improves when protein is adequate and training stimulus is maintained.

Domain 2: Nutrient Protocols & Supplementation

6. Protein: Total Intake, Distribution, and Timing

6.1 Total Daily Protein Targets

Table 6.1 Protein Recommendations by Goal

GOAL	PROTEIN (G/KG/DAY)	EVIDENCE BASE
Hypertrophy / maintenance	1.6–2.2	Morton et al. (2018)
Fat loss (caloric deficit)	2.0–2.4	Pasiakos et al. (2015)
Plant-based athlete	1.8–2.6	Reduced leucine content, lower digestibility

Older adult (60+)	1.8–2.4	Anabolic resistance, higher leucine threshold
Gentle entry / ED risk	1.2–1.6	Avoid fixation; focus on addition, not restriction

6.2 Per-Meal Dosing: The Leucine Distribution Model

Total protein must be distributed across meals to achieve the leucine threshold at each feeding. The optimal pattern is 3–5 meals at 0.4–0.55 g/kg/meal. Skewed distribution (e.g., 70% of daily protein in one meal) produces inferior MPS over 24 hours compared to even distribution, even when total intake is equated.

6.3 Pre-Sleep Protein

Casein (30–50 g) consumed 30–60 minutes before sleep elevates MPS throughout the nocturnal fast, improves overnight whole-body protein balance, and augments the adaptive response to training. The gel-formation property of casein produces a sustained 4–6 hour amino acid release profile, superior to whey's rapid spike-and-decline kinetics for this specific context. This is particularly relevant during caloric deficit or in older adults.

6.4 Protein Quality and Leucine Content

Table 6.2 Leucine Content of Common Protein Sources (per 100 g)

SOURCE	PROTEIN (G)	LEUCINE (G)	LEUCINE PER 25 G PROTEIN
Whey protein isolate	90	10.9	3.0 g
Chicken breast	31	2.6	2.1 g
Lean beef	26	2.1	2.0 g
Eggs (whole, 100 g)	13	1.1	2.1 g
Greek yogurt	10	0.9	2.3 g
Tofu (firm)	8	0.6	1.9 g
Tempeh	19	1.4	1.8 g
Lentils (cooked)	9	0.6	1.7 g
Soy protein isolate	88	7.5	2.1 g

CHAPTER SUMMARY

Total daily protein intake is the highest-priority variable, with targets of 1.6–2.2 g/kg for maintenance and 2.0–2.4 g/kg during caloric deficit. Distribution across meals is the secondary priority: aim for 0.4–0.55 g/kg per meal across 3–5 feedings to hit the leucine threshold at each meal. Pre-sleep casein (30–50 g) provides a sustained amino acid release throughout the nocturnal fast, making it a useful tool during deficits or in older adults. Timing beyond pre-sleep is a distant third priority — total intake and leucine distribution drive the majority of the adaptive response.

◈ MAKE THIS WORK FOR YOU

Your practical protein strategy depends on your schedule, appetite, and digestion. A client who struggles with morning appetite cannot force a 40 g breakfast; instead, they can distribute their intake across 3 well-sized meals (lunch, dinner, and a substantial pre-sleep snack). The principle is flexible, but the leucine threshold per meal is not negotiable.

If you have a small appetite or low calorie budget (e.g., a smaller female in a deficit): Prioritise protein density — choose lean meats, egg whites, and protein isolates that deliver high protein per calorie. Avoid protein sources that come bundled with significant fat or carbohydrates unless they fit your remaining macros.

If you are plant-based: Increase your per-meal protein target by 15–25% to compensate for lower leucine density. Where an omnivore might need 25 g of protein per meal to hit the leucine threshold, you may need 30–35 g. Include at least one serving of soy (tofu, tempeh, edamame) daily as it has the most favourable amino acid profile among plant proteins.

If you are over 50: Anabolic resistance is real, but manageable. Your per-meal protein target is 30–40 g, and your total should be at the upper end of the range (1.8–2.4 g/kg/day). Pre-sleep protein is particularly valuable for you because the nocturnal fast is longer relative to MPS duration.

7. Carbohydrates: Timing, Periodisation, and Compartments

7.1 Carbohydrate Recommendations by Goal

Table 7.1 Carbohydrate Intake by Activity Level and Goal

CONTEXT	G/KG/DAY	NOTES
Strength athlete (maintenance)	4–6	Training volume-dependent; higher on leg days
General hypertrophy	3–5	Moderate; adjust based on recovery and performance
Fat loss (low-moderate activity)	2–3	Prioritise around training window
Fat loss (high activity)	2.5–4	Preserve training performance; deficit in non-training meals
Ketogenic / very low carb	<0.5	Context-dependent; requires electrolyte management

7.2 Training-Day Carbohydrate Periodisation

Strategic carbohydrate timing follows the principle of "training demand, not arbitrary schedule." On training days, concentrate carbohydrates in the pre-training, intra-training, and post-training windows to maximise performance and glycogen resynthesis. On rest days, reduce carbohydrate load to improve insulin sensitivity and reliance on fat oxidation. This does not mean zero carbohydrate on rest days; it means reducing the amount and prioritising lower-glycaemic sources.

7.3 The Post-Training Glycogen Window

Muscle glycogen resynthesis occurs most rapidly in the 30–120 minute post-training window, driven by contraction-mediated GLUT4 translocation (which remains elevated for 24–48 hours). High-glycaemic carbohydrates (0.8–1.2 g/kg immediately post-training, then every 2 hours) optimise depletion-driven glycogen resynthesis. Adding protein (0.3–0.4 g/kg) to this window enhances glycogen storage through insulin potentiation in addition to supporting MPS.

7.4 CPT1 and Mitochondrial Fat Oxidation

The carnitine palmitoyltransferase 1 (CPT1) system controls the rate-limiting step for mitochondrial fatty acid oxidation. Malonyl-CoA, produced from glucose metabolism, inhibits

CPT1—creating the carbohydrate-fat crosstalk: when carbohydrate availability is high, fat oxidation is suppressed. Practical application: training in low-glycogen conditions (e.g., fasted morning cardio, or the final sets of a high-volume session) shifts reliance toward fat oxidation through reduced malonyl-CoA inhibition.

CHAPTER SUMMARY

Carbohydrate intake should be periodised around training demand: higher on training days to support performance and glycogen resynthesis, lower on rest days to improve insulin sensitivity. The post-training glycogen window (30–120 minutes) benefits from 0.8–1.2 g/kg of high-glycaemic carbohydrates, with added protein for enhanced glycogen storage and MPS support. Malonyl-CoA inhibition of CPT1 explains why fat oxidation is suppressed when carbohydrate availability is high. The key coaching takeaway is that carbohydrates are not to be feared — they are performance-supporting nutrients that should align intake with energy expenditure.

◈ MAKE THIS WORK FOR YOU

Your carbohydrate periodisation should reflect your training schedule, not a rigid template. If you train in the evening after work, your pre-training meal (lunch) should be the most carbohydrate-dense meal of the day, and your post-training dinner should complete glycogen resynthesis. If you train first thing in the morning, a smaller pre-training snack may suffice, and your post-training breakfast becomes the priority.

If you train early morning fasted: Your glycogen stores are naturally lower after the overnight fast. Training quality may suffer, especially for high-volume or high-intensity sessions. Consider at least a small pre-training carbohydrate (15–30 g, e.g., a banana or a rice cake with jam) to maintain performance without causing digestive discomfort.

If you are highly active (6+ sessions per week): Your rest-day carbohydrate reduction should be modest — dropping from 5 g/kg to 3–4 g/kg rather than to 2 g/kg. Your training frequency means that even on rest days, glycogen resynthesis for the next training day is a priority. Very low carbohydrate rest days are not appropriate for high-volume athletes.

If you experience energy crashes or poor sleep on low-carb days: Increase your evening carbohydrate slightly. The relationship between carbohydrate intake, cortisol, and sleep quality is individual — some people need a moderate amount of carbohydrate in their last meal for optimal sleep onset and depth.

8. Fats: Essential Fatty Acids and Meal Design

8.1 Fat Recommendations by Goal

Table 8.1 Dietary Fat Intake Guidelines

CONTEXT	G/KG/DAY	MINIMUM
Maintenance / hypertrophy	0.8–1.2	0.5 g/kg/day
Fat loss	0.6–1.0	0.5 g/kg/day
Female athlete	0.8–1.2	0.6 g/kg/day (higher minimum due to hormonal considerations)
Low-carb / ketogenic	1.5–2.5	Fill remaining calories after protein allocation

8.2 Fat-Soluble Vitamin Absorption

Dietary fat is required for the absorption of vitamins A, D, E, and K. Clients on very-low-fat diets (<20% of calories from fat) risk deficiency in these vitamins regardless of intake. Including a source of fat (at least 5–10 g) with meals containing fat-soluble vitamins or carotenoid-rich vegetables significantly improves absorption.

CHAPTER SUMMARY

Fat intake should meet a minimum of 0.5 g/kg/day (0.6 g/kg for female athletes) to support hormonal health, with higher intakes of 0.8–1.2 g/kg for maintenance or hypertrophy phases. The minimum thresholds are non-negotiable — dropping below them compromises testosterone production, menstrual cycle regularity, and fat-soluble vitamin absorption. Beyond the minimum, fat quality (unsaturated over saturated) is more important than precise quantity. Clients on very-low-fat diets risk deficiencies in vitamins A, D, E, and K regardless of intake.

◈ MAKE THIS WORK FOR YOU

Fat intake is not just about hitting a number — it is about the composition, distribution, and context of your meals. A client who drizzles olive oil over roasted vegetables and adds avocado to their salad will have a vastly different inflammatory profile from someone who gets most of their fat from processed sources, even if the total grams are identical.

If you struggle to digest high-fat meals: Distribute fat more evenly across the day rather than concentrating it in one meal. Fat slows gastric emptying significantly — a 40 g fat meal will sit heavier than four 10 g fat meals. If you train soon after eating, keep the pre-training meal lower in fat (under 15 g).

If you are in a fat loss phase and fat intake is low: Use EPA/DHA supplementation (1.5–2 g/day) as insurance against the inflammatory effects of a low-fat diet. When fat intake drops below 0.6 g/kg/day, the omega-3 to omega-6 ratio tends to worsen because the fats that remain are often from lean animal sources with minimal omega-3 content.

If you follow a high-carb, low-fat approach: Ensure your training performance and recovery are not compromised. Some individuals thrive on low fat (0.3–0.5 g/kg/day) for short periods, but after 8–12 weeks, hormonal markers (libido, sleep quality, menstrual regularity) tend to decline. Monitor these subjective markers closely and increase fat at the first sign of deterioration.

9. Vitamin D & Calcium

9.1 Physiological Roles

Vitamin D functions as a steroid hormone with receptors throughout the body. Beyond calcium homeostasis, it modulates testosterone synthesis (via upregulation of CYP11A and CYP17A in Leydig cells), immune function (regulating antimicrobial peptide expression), glucose metabolism (improving insulin sensitivity through VDR-mediated transcription), and inflammation through NF-kB pathway suppression. The vitamin D receptor (VDR) is expressed in skeletal muscle tissue, where it regulates myoblast proliferation and differentiation.

9.2 Athlete-Specific Protocol

Table 9.1 Vitamin D Protocol for Athletes

PARAMETER	RECOMMENDATION
Testing	Serum 25(OH)D; optimal range 50–80 ng/mL
General maintenance	1,000–2,000 IU/day
Deficiency correction	5,000 IU/day or 50,000 IU/week for 8 weeks
Food sources	Fatty fish (salmon 600 IU/100g), egg yolk (40 IU), UV-exposed mushrooms
Co-factors	Vitamin K2 (100–200 mcg/day), magnesium (to activate VDR)

9.3 Calcium

While vitamin D controls calcium absorption, calcium itself plays direct roles in muscle contraction (excitation-contraction coupling via troponin C), bone mineral density (peak bone mass development is critical in adolescence and early adulthood), and fat metabolism (dietary calcium modulates adipocyte lipid metabolism through calcitriol suppression). Athlete RDA: 1,000–1,300 mg/day, with dairy, fortified plant milks, leafy greens, and tinned fish (with bones) as primary sources.

CHAPTER SUMMARY

Vitamin D is the most commonly indicated supplement for athletes, with an optimal serum range of 50–80 ng/mL (25(OH)D). Testing before high-dose supplementation is ideal, but 1,000–2,000 IU/day is safe as a general maintenance dose for most adults. Vitamin D receptors are expressed in skeletal muscle, where they regulate myoblast proliferation and differentiation, linking vitamin D status directly to muscle health. Calcium (1,000–1,300 mg/day) supports muscle contraction, bone mineral density, and fat metabolism, with dairy and fortified plant milks as the primary sources.

◈ MAKE THIS WORK FOR YOU

Vitamin D requirements are highly individual and depend on your latitude, skin tone, sun exposure habits, and body weight. A fair-skinned individual who spends 15 minutes outside at midday in summer may produce 10,000–20,000 IU of vitamin D endogenously; someone with darker skin living at a northern latitude in winter may produce virtually none despite spending the same time outside.

If you live in a region with limited winter sun (north of 35° latitude, or anywhere with significant cloud cover): Supplementing at 1,000–2,000 IU/day during autumn and winter is a reasonable default. If you have risk factors for deficiency (BMI >30, limited outdoor time, darker skin, or gut malabsorption), consider having your 25(OH)D tested and dosing accordingly.

If you are vegan or vegetarian: Your calcium intake requires attention if you do not consume dairy. Fortified plant milks (300–450 mg calcium per serving), calcium-set tofu, and leafy greens are your primary sources. Aim for at least 3 servings of calcium-rich foods per day. For vitamin D, choose a vegan D3 supplement derived from lichen rather than the less effective D2 form.

Vitamin D and K2 synergy: If you supplement vitamin D at 1,000+ IU/day, consider adding vitamin K2 (100–200 mcg/day) to direct calcium toward bone rather than soft tissues. This is especially relevant if your calcium intake is also supplemented.

10. Magnesium

10.1 Physiological Roles

Magnesium is a required co-factor for over 300 enzymatic reactions. In the athletic context, it is critical for: ATP production (Mg-ATP complex is the biologically active form), GLUT4 translocation (insulin signalling requires magnesium), muscle relaxation (antagonises calcium at the sarcomere, preventing cramping), HPA axis regulation (modulates cortisol clearance), GABA modulation (promotes sleep quality through NMDA receptor antagonism), and testosterone synthesis (involved in the steroidogenic pathway).

10.2 Athlete Protocol

Table 10.1 Magnesium Supplementation Guide

FORM	ABSORPTION	BEST FOR	DOSE
Magnesium glycinate	High	Sleep, stress, general deficiency	200-400 mg elemental Mg
Magnesium malate	High	Energy, muscle pain	200-400 mg elemental Mg
Magnesium citrate	Moderate	General use, constipation	200-400 mg elemental Mg
Magnesium oxide	Low	Avoid — poorly absorbed	—
Magnesium L-threonate	High	Cognitive effects, BBB penetration	144-200 mg elemental Mg

10.3 Deficiency Screening

Clinical magnesium deficiency is underdiagnosed because serum magnesium is tightly regulated and does not reflect total body stores. Red flags include: nocturnal muscle cramps, restless legs, anxiety/sleep disruption, insulin resistance, and chronic constipation. During caloric deficit or high training volume, magnesium requirements increase. Food sources: pumpkin seeds (262 mg/30 g), almonds (80 mg/30 g), spinach (87 mg/cup cooked), dark chocolate (64 mg/30 g).

CHAPTER SUMMARY

Magnesium is a co-factor for over 300 enzymatic reactions and is critical for ATP production, GLUT4 translocation, muscle relaxation, sleep quality, and testosterone synthesis. Magnesium glycinate (200–400 mg elemental Mg before bed) is the most versatile form, ideal for clients reporting sleep disruption, anxiety, nocturnal cramps, or high training volume. Serum magnesium is unreliable; RBC magnesium testing provides a more accurate reflection of total body stores. Athletes in caloric deficit or under high training volume are at elevated risk of subclinical deficiency.

◈ MAKE THIS WORK FOR YOU

Choosing the right magnesium form and dose is a personal decision that depends on your primary symptoms and goals. Magnesium glycinate is the best choice if sleep quality, anxiety, or stress are your main concerns. Magnesium malate may be preferable if you are taking it for energy production or muscle pain. Magnesium citrate is an adequate general option but has a mild laxative effect at higher doses.

If you experience nocturnal muscle cramps or restless legs: Magnesium glycinate 200–400 mg before bed is the most evidence-backed intervention in this context. Combine it with adequate hydration and electrolyte balance (especially sodium and potassium) for comprehensive cramp prevention. Cramps are rarely caused by magnesium alone.

If you are in a caloric deficit or have high training volume: Your magnesium requirements are elevated because sweat losses increase and dietary intake tends to drop with lower overall food consumption. This is the most common scenario where subclinical magnesium deficiency develops. Consider 200–400 mg of magnesium glycinate as a preventative measure even if you do not have obvious symptoms.

If you have gut issues or malabsorption (IBD, celiac, chronic diarrhoea): You are at elevated risk of deficiency regardless of intake. RBC magnesium testing is more informative than serum magnesium in your case. If supplementation causes digestive upset, try magnesium glycinate (most gentle on the stomach) and take it with food.

11. Zinc & Iron

11.1 Zinc

Zinc plays a foundational role in testosterone synthesis (required for the steroidogenic acute regulatory protein and 17-hydroxylase activities), IGF-1 signalling (zinc finger proteins are essential for IGF-1 receptor function), antioxidant defence (co-factor for superoxide dismutase), protein synthesis (required for ribosomal function and mRNA stability), and immune function (T-cell maturation and function). Athlete RDA: 11–15 mg/day. Overtraining and high sweat losses increase requirements.

11.2 Iron

Iron is essential for oxygen transport (haemoglobin), aerobic energy production (cytochrome c oxidase in the electron transport chain), myoglobin synthesis (oxygen storage in muscle), and neurotransmitter synthesis. Athlete requirements are elevated due to foot-strike haemolysis, sweat losses, GI bleeding, and increased erythropoiesis. Female athletes are at particular risk due to menstrual losses (additional 1–2 mg/day).

Critical: Iron Supplementation Requires Testing

Iron supplementation without confirmed deficiency is dangerous due to the risk of iron overload (haemochromatosis). Test ferritin (optimal 50–150 ng/mL for athletes), serum iron, TIBC, and transferrin saturation before supplementing. Iron absorption is enhanced by vitamin C and inhibited by calcium, tannins (tea), and phytates (grains, legumes).

11.3 Zinc-Iron-Copper Interaction

Zinc and iron compete for absorption through the divalent metal transporter (DMT1). High-dose zinc supplementation (>40 mg/day) can suppress copper absorption, leading to copper deficiency. When supplementing zinc long-term (>3 months), include 1–2 mg copper per 15–30 mg zinc. Iron should be taken separately from zinc and calcium (morning and evening, or with different meals).

CHAPTER SUMMARY

Zinc supports testosterone synthesis, IGF-1 signalling, and immune function, with an athlete RDA of 11–15 mg/day. Iron is essential for oxygen transport and aerobic energy production, with female athletes at particular risk due to menstrual losses. Iron supplementation must not be initiated without testing ferritin and full iron studies first, as iron overload (haemochromatosis) is a serious risk. Long-term zinc supplementation (>40 mg/day for >3 months) requires copper co-supplementation (1–2 mg copper per 15–30 mg zinc) due to DMT1 transport competition.

◈ MAKE THIS WORK FOR YOU

Zinc and iron status are strongly influenced by your diet type and lifestyle. A vegetarian or vegan athlete faces a double risk: lower intake of both minerals and reduced absorption due to phytates and the absence of haem iron. A female athlete with heavy menstrual losses may need more iron than standard recommendations. A male athlete consuming adequate red meat likely meets both requirements without supplementation.

If you are a vegetarian or vegan: Pay deliberate attention to zinc and iron intake. For zinc, soak and sprout legumes and grains to reduce phytate content, and include at least one serving of pumpkin seeds, lentils, or cashews daily. For iron, pair iron-rich plant foods (spinach, lentils, fortified cereals) with vitamin C (citrus, bell peppers) to enhance absorption, and avoid tea or coffee within one hour of iron-rich meals.

If you are a female athlete with heavy periods: You are at the highest risk of iron deficiency in the athletic population. Have your ferritin checked annually as a baseline. Do not self-prescribe iron without testing — the symptoms of iron deficiency (fatigue, pallor, exercise intolerance) overlap with many other conditions, and iron overload carries its own risks.

If you supplement zinc long-term: Any dose above 15 mg/day sustained for more than 3 months should be accompanied by 1–2 mg of copper to prevent copper deficiency. This is a commonly missed interaction. Take zinc with a protein-containing meal to reduce gastric discomfort, and separate it from iron and calcium supplements by at least 4 hours.

12. B-Complex & Energy Metabolism

The B-complex vitamins function primarily as co-enzymes in energy metabolism. Each plays a specific role in the metabolic pathways that convert dietary energy into usable ATP:

Table 12.1 B-Vitamin Functions in Athletic Context

VITAMIN	ACTIVE FORM	METABOLIC ROLE	ATHLETE RDA
B1 (Thiamine)	TPP	Pyruvate dehydrogenase (glucose to acetyl-CoA), branched-chain amino acid metabolism	1.5-2.0 mg
B2 (Riboflavin)	FAD/FMN	Electron transport chain (Complex I and II), fatty acid oxidation	1.5-2.0 mg
B3 (Niacin)	NAD/NADP	Redox reactions in glycolysis, TCA cycle, electron transport chain	16-20 mg
B5 (Pantothenic acid)	CoA	Acetyl-CoA formation (TCA cycle entry), fatty acid synthesis/oxidation	5-10 mg
B6 (Pyridoxine)	PLP	Glycogen phosphorylase (glycogen breakdown), amino acid transamination	2-3 mg
B7 (Biotin)	—	Carboxylation reactions in gluconeogenesis, fatty acid synthesis	30-100 mcg
B9 (Folate)	THF	Amino acid metabolism, red blood cell production, homocysteine regulation	400-800 mcg DFE
B12 (Cobalamin)	MeCbl, AdoCbl	Methionine synthesis (methylation), succinyl-CoA (fatty acid metabolism)	2.5-5.0 mcg

During caloric deficit or high-volume training, B-vitamin requirements increase because energy flux through these pathways is elevated relative to intake. Vegans and vegetarians are at particular risk for B12 deficiency, which requires supplementation regardless of diet quality.

CHAPTER SUMMARY

B-vitamins are co-enzymes in energy metabolism. Vegans need B12 supplementation. Caloric deficit increases B-vitamin requirements. Food-first approach before supplementing.

◈ MAKE THIS WORK FOR YOU

B-vitamin needs are individualised by diet, training volume, and stress levels. The most important personalised recommendation is straightforward: if you follow a vegan diet, supplement B12. This is non-negotiable — no plant food provides adequate B12, and deficiency causes irreversible neurological damage over time.

If you are in a caloric deficit or have high training volume: Your B-vitamin requirements increase because energy flux through metabolic pathways is elevated while dietary intake is reduced. This is a scenario where a B-complex supplement (providing 100–200% of the RDA for each B-vitamin) may be prudent as insurance, especially if your diet is not rich in whole grains, leafy greens, and animal products.

If you have a genetic MTHFR variant (common, affecting 30–60% of the population): Your ability to convert folic acid to its active form (5-MTHF) is reduced. If you supplement B9, choose the methylated form (L-methylfolate or 5-MTHF) rather than folic acid. This is especially relevant if you are planning pregnancy, as folate status is critical for neural tube development.

Practical tip: B-vitamins are water-soluble, so toxicity is rarely a concern, but timing matters — taking B-vitamins in the evening can disrupt sleep for some individuals due to their role in energy metabolism. Morning or early afternoon is preferable.

13. Electrolytes & Hydration

13.1 Daily Hydration Targets

Table 13.1 Hydration Requirements by Activity Level

ACTIVITY LEVEL	DAILY WATER TARGET (L)
Sedentary	2.0–2.5
Moderately active (3–5 hrs/week)	2.5–3.5
Very active (6–10 hrs/week)	3.5–5.0
Extreme (10+ hrs/week, outdoor heat)	5.0–7.0+

13.2 Training Hydration Protocol

Pre-training (2 hours before): 500–600 mL water. **Intra-training**: 200–300 mL every 15–20 minutes. Sessions exceeding 60 minutes in warm conditions require electrolyte replacement: 300–600 mg sodium, 150–300 mg potassium, 30–60 mg magnesium per litre of fluid. **Post-training**: 1.25–1.5 L per kg of body weight lost during exercise.

13.3 Hydration Status Assessment

Urine colour is a practical daily screening tool: pale straw indicates euhydration; dark amber (reminiscent of apple juice) indicates dehydration and requires immediate increased fluid intake. Additional indicators: thirst (a late signal; do not rely on it), skin turgor, tear production, and cognitive function. A morning body weight decrease of >2% from baseline suggests hypohydration.

13.4 Electrolyte Balance

Each electrolyte plays a distinct role in athletic performance: **Sodium** (3–5 g/day) is the primary extracellular cation, critical for fluid balance and nerve conduction. Losses of 500–1,500 mg per hour of intense exercise are typical. **Potassium** (3.5–5 g/day) is the primary intracellular cation, involved in muscle contraction and glycogen storage. **Calcium** (1,000–1,300 mg/day) triggers muscle contraction through excitation-contraction coupling. **Magnesium** (400–600 mg/day for athletes) is required for muscle relaxation and ATP function.

CHAPTER SUMMARY

Hydration targets scale with activity. Electrolyte replacement matters for sessions >60 min. Urine colour is the best practical screening tool. Sodium is the primary electrolyte lost in sweat.

◈ MAKE THIS WORK FOR YOU

Your individual hydration needs depend on your body size, sweat rate, training environment, and acclimatisation status. A 90 kg individual training in a humid gym in summer loses 1–2 L of sweat per hour, while a 60 kg individual doing the same session in an air-conditioned environment may lose 0.5–1 L. The "2 L per day" rule is meaningless for both — one is chronically dehydrated, the other is fine.

Calculate your sweat rate: Weigh yourself naked before and after a training session. Every kilogram of weight lost equals approximately 1 L of fluid deficit. Your goal is to keep this deficit under 2% of body weight (e.g., under 1.4 kg for a 70 kg person). This gives you a personalised hydration target for future sessions.

If you are a heavy sweater or train in hot conditions: Sodium replacement is your priority. A pinch of salt (1–2 g) in your pre-training water, or an electrolyte tab with 300–600 mg sodium during longer sessions, can make a noticeable difference to cramping, energy, and focus. Plain water alone, in large volumes, can actually dilute blood sodium (hyponatremia) in prolonged sessions.

If you are prone to frequent urination at night: Front-load your hydration — drink most of your daily water before 6 PM, and taper off in the evening. Your total daily target is the same, but shifting the timing eliminates sleep disruption without compromising hydration status.

14. Antioxidant Network & Polyphenols

14.1 The Nrf2-Keap1 System

The nuclear factor erythroid 2-related factor 2 (Nrf2) is the master regulator of the antioxidant response. Under baseline conditions, Nrf2 is bound to Kelch-like ECH-associated protein 1 (Keap1) in the cytoplasm, targeting it for degradation. Oxidative stress or Nrf2-activating compounds cause Keap1 dissociation, allowing Nrf2 to translocate to the nucleus and activate antioxidant response elements (AREs), upregulating glutathione peroxidase (GPX), superoxide dismutase (SOD), catalase, and haem oxygenase-1 (HO-1).

14.2 The Training Adaptation Paradox

Exercise-induced reactive oxygen species (ROS) are necessary signalling molecules for training adaptations, including mitochondrial biogenesis and antioxidant enzyme upregulation. High-dose antioxidant supplementation (particularly vitamins C and E) blunts or abolishes these adaptations. The practical solution is food-first antioxidant intake: whole foods provide co-ordinated antioxidant networks at physiological doses that do not suppress training adaptations, unlike megadose supplements.

14.3 Food Sources of Nrf2 Activators

Sulforaphane (broccoli sprouts, broccoli), curcumin (turmeric, requires piperine for absorption), EGCG (green tea), resveratrol (grapes, red wine), quercetin (onions, apples, berries), and hydroxytyrosol (olive oil) activate Nrf2 signalling. Timing these foods post-training may optimise the balance between adaptive signalling and recovery.

14.4 Vitamins C and E

Despite their reputation as antioxidants, high-dose vitamin C (1,000+ mg/day) and vitamin E (400+ IU/day) supplements have been shown to attenuate the training-induced increase in VO2max, insulin sensitivity, and muscle protein synthesis. This does not mean these vitamins are harmful; it means that food sources deliver them in the context of the full phytochemical network, at doses that do not excessively suppress training adaptations.

CHAPTER SUMMARY

Exercise-induced ROS are necessary for training adaptations. High-dose antioxidant supplements (C, E) blunt adaptations. Food-first antioxidant intake from whole foods is superior. Nrf2 activators like sulforaphane support the antioxidant network without suppressing adaptation.

◈ MAKE THIS WORK FOR YOU

Your antioxidant strategy should adapt to your training phase and goals. During a high-volume training block, endogenous antioxidant capacity may be overwhelmed, increasing the need for polyphenol-rich foods. During a competition prep or fat loss phase where calorie intake is lower, antioxidant density per calorie becomes important — prioritise low-calorie, high-antioxidant foods like berries, cruciferous vegetables, and green tea.

If you take high-dose vitamin C (1,000+ mg/day) as an immune supplement: Consider shifting your dose to post-training only, or better yet, getting your vitamin C from food sources (bell peppers, citrus, kiwi, broccoli) at physiological doses. The evidence that megadose antioxidants blunt training adaptations is strongest for vitamin C and E supplements taken near training sessions.

Practical Nrf2 activation: Eating broccoli sprouts or cruciferous vegetables provides sulforaphane, one of the most potent natural Nrf2 activators. Timing these foods post-training (as part of your recovery meal) may optimise the balance between allowing sufficient ROS signalling for adaptation while supporting the antioxidant network for recovery.

If you consume green tea or coffee regularly: The polyphenols (EGCG in green tea, chlorogenic acid in coffee) contribute to your antioxidant network. However, tannins in tea and coffee can inhibit iron absorption — if you are at risk of iron deficiency, consume these beverages at least one hour away from iron-rich meals.

15. Tier 1: High-Confidence Supplements

Tier 1 supplements have consistent, reproducible evidence across multiple meta-analyses demonstrating efficacy for body composition or performance outcomes with a favourable safety profile.

15.1 Creatine Monohydrate

Table 15.1 Creatine Protocol

PARAMETER	RECOMMENDATION
Dosing (loading)	20 g/day (4 x 5 g) for 5-7 days
Dosing (maintenance)	3-5 g/day indefinitely
Form	Creatine monohydrate (not HCl, ethyl ester, or other forms)
Timing	Post-training (insulin-dependent uptake); any time if daily
Onset	5-7 days (loading) or 3-4 weeks (maintenance only)
Safety	Extensive record; safe at recommended doses for long-term use

15.2 Protein Supplements

Whey protein concentrate (WPC) is the standard for general use because it is cost-effective with a complete amino acid profile and high leucine content. Whey protein isolate (WPI) removes more lactose and fat, suitable for those with lactose sensitivity. Casein is preferred for pre-sleep due to its sustained-release profile. Plant-based blends (pea + rice, or soy isolate) can achieve equivalent amino acid profiles when the total dose is increased by 15–25% to compensate for reduced leucine content and digestibility. Third-party testing (Informed Sport, NSF Certified for Sport) is recommended.

15.3 Omega-3 Fatty Acids (EPA/DHA)

Table 15.2 Omega-3 Protocol

PARAMETER	RECOMMENDATION
Dosing (general)	1.5-2.0 g/day EPA+DHA
Dosing (therapeutic, high inflammation)	3-4 g/day EPA+DHA
Form preference	rTG (re-esterified triglyceride) for best absorption
Ratio	EPA:DHA should be at least 1.5:1
Timing	With a fat-containing meal for absorption
Onset	4-12 weeks for membrane incorporation

Safety	Mild blood thinning; discontinue 1 week prior to surgery

15.4 Vitamin D

Already covered in detail in Chapter 9. Supplementation is strongly recommended during winter months or for individuals with limited sun exposure. Testing before supplementation is ideal but not always practical; 1,000–2,000 IU/day as a maintenance dose is safe for almost all adults.

15.5 Magnesium

Already covered in detail in Chapter 10. Supplementation with magnesium glycinate (200–400 mg elemental Mg before bed) is recommended for clients reporting sleep disruption, anxiety, nocturnal cramps, or high training volume. Athletes in caloric deficit are at elevated risk of subclinical deficiency.

CHAPTER SUMMARY

Creatine monohydrate (3-5 g/day) has the strongest evidence of any supplement. Protein supplements are convenient but food is equivalent. Omega-3s (1.5-2 g EPA+DHA/day) support inflammation and recovery. Vitamin D and magnesium are Tier 1 due to widespread deficiency risk.

◈ MAKE THIS WORK FOR YOU

Tier 1 supplements are the closest thing to universal recommendations, but there are individual considerations for each. Creatine monohydrate is effective for almost everyone who performs resistance or high-intensity training, but the response is more noticeable in vegetarians (who start with lower muscle creatine stores by 20–40%). Omega-3 status is highly dependent on your diet — if you eat fatty fish 2+ times per week, you may not need a supplement.

Creatine: The timing, form, and loading phase matter less than consistency. The single most important factor is taking it daily. Five grams of creatine monohydrate taken any time of day produces the same result as a meticulously timed, loaded protocol, provided it is taken consistently over 4+ weeks. Mix it in warm water or coffee for better dissolution.

Omega-3: If you decide to supplement, choose a product that lists the EPA and DHA content specifically, not just "fish oil" or "omega-3s." The therapeutic target is 1.5–2 g of EPA+DHA combined per day. A standard fish oil capsule provides only 300–400 mg, meaning most people need 4–6 capsules to reach the target. Look for concentrated (1,000+ mg EPA+DHA per capsule) products to reduce pill burden.

Vitamin D and magnesium: These are Tier 1 because the prevalence of suboptimal status in the general population and athletic population is high, not because every individual needs them. If you have confirmed adequate status (25(OH)D >50 ng/mL, no symptoms of low magnesium), supplementation may provide minimal benefit. The decision should be based on risk assessment, not blind recommendation.

16. Tier 2: Conditionally Effective Supplements

16.1 Caffeine

Table 16.1 Caffeine Protocol

PARAMETER	RECOMMENDATION
Dosing	3-6 mg/kg body weight, 30-60 minutes pre-training
Form	Anhydrous caffeine (powder/capsule) or black coffee (1-2 cups)
Genetics	CYP1A2 slow metabolisers (approximately 50% of population) benefit less and experience more side effects
Tolerance	Significant; cycle 2 weeks on / 1 week off or reduce daily intake
Stack	Synergistic with 200-400 mg L-theanine (reduces jitters, improves focus)
Sleep impact	Avoid after 2-4 PM depending on individual sensitivity; half-life 3-7 hours

16.2 Beta-Alanine

Mechanism: Carnosine synthesis, which buffers hydrogen ions (H+) during high-intensity exercise, delaying muscular fatigue. Dosing: 4.8–6.4 g/day for 4–6 weeks (loading), then 1.2–2.4 g/day maintenance. Side effect: harmless paresthesia (skin tingling) due to Mas-related G-protein coupled receptor member D (MRGPRD) activation; mitigated by split dosing (1.6 g doses across 3–4 meals) or sustained-release formulations. Effective for 1–4 minute high-intensity efforts. Stacked with creatine for additive benefits.

16.3 Citrulline Malate

Mechanism: Nitric oxide precursor via arginine conversion, enhancing vasodilation, blood flow, and ammonia clearance. Dosing: 6–8 g, 60–90 minutes pre-training. Effective for increasing repetitions in subsequent sets (especially in high-volume training) and reducing perceived exertion. Less effective in well-trained individuals with already-efficient blood flow. Stacked with creatine and beta-alanine for comprehensive training supplementation.

16.4 Casein (Pre-Sleep)

Covered in Chapter 6.3. The protocol: 30–50 g casein 30–60 minutes before sleep. Options: micellar casein powder, cottage cheese, Greek yogurt (strained), or a protein blend with casein-dominant profile.

CHAPTER SUMMARY

Caffeine (3-6 mg/kg pre-training) is effective but tolerance develops. Beta-alanine (4.8-6.4 g/day loading) buffers H+ for high-intensity work. Citrulline malate (6-8 g pre-training) improves blood flow. Casein pre-sleep (30-50 g) supports overnight MPS.

◈ MAKE THIS WORK FOR YOU

Tier 2 supplements should be selected based on your specific training demands, not used as a generic stack. If you do not perform high-intensity efforts lasting 1–4 minutes (the domain where beta-alanine's H+ buffering matters), beta-alanine will not improve your training outcomes. If you do not experience noticeable pumps or performance drop-off in higher rep ranges, citrulline malate may not be worth the cost.

Caffeine: Your CYP1A2 genotype determines whether caffeine is ergogenic for you or simply increases anxiety without performance benefit. Approximately 50% of the population are slow metabolisers. If caffeine makes you jittery, anxious, or disrupts your sleep even when taken early, you may be a slow metaboliser — consider reducing your dose or avoiding pre-training caffeine entirely. For fast metabolisers, 3–6 mg/kg 30–60 minutes pre-training is effective.

Beta-alanine: The 4.8–6.4 g/day loading phase causes harmless paresthesia (skin tingling) in most people. If you find this unpleasant, split the daily dose into 1.6 g servings taken every 3–4 hours, or use a sustained-release formulation. The tingling is not dangerous and diminishes as you habituate, but if it interferes with your training focus, split dosing is an effective workaround.

Citrulline malate: 6–8 g taken 60–90 minutes before training on an empty stomach (or with a very small snack) produces the most consistent effects. The benefit is most noticeable in the later sets of a high-volume session — if your training is primarily low-rep, heavy strength work, the effect may be negligible.

17. Tier 3: Emerging & Contextual Supplements

17.1 Ashwagandha (Withania somnifera)

Mechanism: HPA axis modulation via cortisol reduction. Dosing: KSM-66 extract 300–600 mg/day, or Sensoril 125–250 mg/day. Cycling: 8–12 weeks on, 2–4 weeks off. Effective primarily for individuals with elevated baseline cortisol (high stress, sleep deprivation, overtraining). Stack with magnesium glycinate. Evidence tier: moderate for stress reduction; weak for direct muscle gain or fat loss.

17.2 HMB (Beta-Hydroxy Beta-Methylbutyrate)

A leucine metabolite that inhibits ubiquitin-proteasome proteolysis and may activate mTOR. Dosing: 3 g/day (1 g with three meals). Context-dependent: most effective in untrained individuals initiating training, in caloric deficit for preserving lean mass, and in older adults with anabolic resistance. Limited benefit in well-trained individuals with adequate protein intake.

17.3 Boron

Mechanism: Reduces sex hormone-binding globulin (SHBG), increasing free testosterone. Dosing: 3–6 mg/day. Evidence tier: Tier 3 exploratory. Small effect sizes; most useful when SHBG is known to be elevated. Food sources: avocados, almonds, raisins, chickpeas. Caution: narrow therapeutic window; avoid exceeding 10 mg/day.

17.4 Supplement Decision Framework

When to Recommend a Supplement

1. **Risk assessment:** Is there a confirmed or likely deficiency? (Vitamin D, Magnesium, B12 for vegans)
2. **Evidence threshold:** Does the supplement have Tier 1 or Tier 2 evidence for the specific goal?
3. **Cost-benefit:** Is the cost justified relative to the expected effect size?
4. **Third-party verification:** Does the product have Informed Sport, NSF, or USP certification?
5. **Baseline diet:** Has diet been optimised first? Supplements do not compensate for poor nutrition.
6. **Stack check:** Are there known interactions with other supplements or medications?

CHAPTER SUMMARY

Ashwagandha reduces cortisol in stressed individuals. HMB is context-dependent (untrained, deficit, elderly). Boron may reduce SHBG. Apply the supplement decision framework: deficiency first, then evidence tier, then cost-benefit, then third-party verification.

◈ MAKE THIS WORK FOR YOU

Tier 3 supplements should be treated as experiments, not staples. For each one, run a personal N=1 trial: use it consistently for 4–8 weeks, track relevant outcomes (subjective stress, recovery, libido, training performance), then decide whether the cost justifies the benefit. If you cannot tell whether a supplement is working after 8 weeks, it probably is not worth continuing.

Ashwagandha: Most effective for individuals with chronically elevated cortisol — those reporting high stress, poor sleep, and difficulty recovering despite adequate nutrition. If your stress management is good and your sleep is solid, ashwagandha is unlikely to produce noticeable effects. Use the KSM-66 extract (300–600 mg/day) for standardised dosing. Cycle 8–12 weeks on, 2–4 weeks off.

HMB: This is not a supplement for the experienced trainee with adequate protein intake. HMB's effects are largest in three specific populations: (1) beginners starting resistance training, (2) individuals in a severe caloric deficit where muscle preservation is challenged, and (3) older adults with anabolic resistance. If you do not fit one of these profiles, spend the money on higher-quality food first.

Boron: The SHBG-lowering effect is real but modest (3–6 mg/day reduces SHBG by approximately 10–15%). This may be relevant if you have symptoms of low free testosterone with normal total testosterone — but the effect is unlikely to be noticeable for most individuals. The therapeutic window is narrow; do not exceed 10 mg/day.

COACHING APPLICATION

Supplement Audit and Macronutrient Restructure

Scenario: A 28-year-old female client, 65 kg, trains 5x/week (3 resistance + 2 HIIT). She reports fatigue, poor recovery, and irritability. She takes vitamin C (1,000 mg), iron (self-prescribed 65 mg), and BCAAs. She eats approximately 1,800 kcal/day with 80 g protein, 220 g carbs, 55 g fat.

Assessment: Protein is severely insufficient (1.2 g/kg vs. 1.6-2.2 target). Iron supplementation without testing is risky. BCAAs provide no advantage over complete protein. Vitamin C at this dose may blunt training adaptations. Omega-3 intake is low (no fish, no supplement).

Intervention: (1) Increase protein to 110-130 g/day (1.7-2.0 g/kg) by redistributing calories from carbs, keeping total intake stable. (2) Replace BCAAs with whey protein — same cost, complete amino acid profile. (3) Stop iron until ferritin is tested. (4) Add omega-3 (2 g EPA+DHA/day) and magnesium glycinate (200 mg before bed). (5) Move vitamin C to post-training only, or get it from food. (6) Reassess energy, recovery, and sleep after 3 weeks.

Key Takeaway: Macronutrient fundamentals (total protein) must be optimised before supplements can provide marginal benefit. Testing before supplementing iron prevents iatrogenic harm. Supplement stacks need regular auditing for interactions and evidence strength.

Domain 3: Applied Protocols & Food Reference

18. Calorie & Macronutrient Calculator

18.1 BMR Estimation (Mifflin-St Jeor)

$$\text{Male: BMR} = 10 \times \text{weight (kg)} + 6.25 \times \text{height (cm)} - 5 \times \text{age (y)} + 5$$

$$\text{Female: BMR} = 10 \times \text{weight (kg)} + 6.25 \times \text{height (cm)} - 5 \times \text{age (y)} - 161$$

18.2 TDEE Activity Multipliers

Table 18.1 Activity Level Adjustments

ACTIVITY LEVEL	MULTIPLIER	DESCRIPTION
Sedentary	1.2	Desk job, minimal movement
Light activity	1.375	1-3 days/week structured exercise
Moderate activity	1.55	3-5 days/week structured exercise
Very active	1.725	6-7 days/week, intense training
Extreme	1.9	Twice-daily training, physical labour

18.3 Calorie Targets by Goal

Table 18.2 Calorie Adjustment by Goal

GOAL	ADJUSTMENT FROM TDEE	RATE OF CHANGE
Fat loss (aggressive)	-25-30%	0.7-1.0% body weight/week
Fat loss (moderate)	-15-25%	0.5-0.7% body weight/week
Recomp (body recomposition)	-100-300 kcal deficit	0.3-0.5% body weight/week (if weight change occurs)
Lean gain / maintenance	TDEE to +5%	<0.5% body weight/month

Muscle gain (aggressive)	+10-20%	0.5-1.5% body weight/month

18.4 Macronutrient Distribution

Table 18.3 Macronutrient Calculation Logic

STEP	CALCULATION
1. Protein	Goal-based g/kg/day (Table 6.1). Multiply by body weight (kg).
2. Fat	Minimum 0.5-0.8 g/kg/day (Table 8.1). Higher for female athletes and low-carb contexts.
3. Carbohydrate	Remaining calories after protein and fat allocation.
4. Adjustment	Reassess after 2-4 weeks based on actual rate of change and subjective feedback.

18.5 Two-Week Adjustment Rule

When the rate of weight change is less than half the target rate over two consecutive weeks, adjust calories by 100–200 kcal/day in the direction of the goal (reduce for faster loss; increase for faster gain). When weight loss exceeds 1% of body weight per week for two consecutive weeks, increase calories by 100–200 kcal/day to avoid excessive metabolic adaptation. Scale weight should be interpreted with context: glycogen depletion, sodium intake, menstrual cycle phase, and creatine loading all cause water weight fluctuations of 1–3 kg.

CHAPTER SUMMARY

Mifflin-St Jeor is the most accurate BMR equation for most populations. Start at 15–25% deficit for fat loss, +10–20% for muscle gain. Protein is set first, then fat minimums, then carbs fill the remainder. The 2-week adjustment rule prevents overreaction to water-weight fluctuations.

◈ MAKE THIS WORK FOR YOU

The numbers from these equations are starting points, not prescriptions. A 35-year-old male who is 85 kg with a desk job but trains 5x/week will not have the same TDEE as an 85 kg construction worker who trains 3x/week, even though both select the same "activity multiplier." The equations are population averages — your individual TDEE is whatever your 3-week average weight trend says it is.

Your personalised starting point: Calculate your estimated TDEE using Mifflin-St Jeor, set your calories to the target for your goal (deficit, maintenance, surplus), then track your weight daily for 2–3 weeks. Use a 7-day rolling average to filter daily fluctuations. If your weight is moving at the expected rate (0.5–1% per week for fat loss), continue. If it is moving faster or slower, adjust by 100–200 kcal/day and re-evaluate.

For macronutrient distribution: Protein and fat have non-negotiable minimums (1.6 g/kg protein, 0.5–0.8 g/kg fat). Carbohydrates fill the remainder. If this leaves you with very low carbs (<2 g/kg), consider whether your training performance is adequately supported. If so, proceed. If not, redistribute calories from fat to carbs while keeping fat at its minimum.

Do not chase an exact number: A range of 2,200–2,500 kcal produces the same results as a fixed 2,350 kcal. The exactness of the calculation is false precision — what matters is consistency of adherence and appropriate adjustment over time.

19. Meal Structure Library

19.1 Meal Structure Templates

Table 19.1 Meal Structure Options

PATTERN	STRUCTURE	BEST FOR
3 meals	Breakfast ~30%, Lunch ~35%, Dinner ~35%	Standard schedule, family meals
4 meals	Meals at 0, 4, 8, 11 hours post-wake	Protein distribution optimisation
2 meals + snacks	2 main meals + flexible snacks	Adherence preference
16:8 IF (2 meals)	Feeding window 12–8 PM, 2 meals	Adherence, insulin sensitivity (context-dependent)

19.2 Sample Meal Templates (per meal, ~700–800 kcal)

Table 19.2 Meal Template Examples

TEMPLATE	COMPONENTS	PROTEIN (G)	CARBS (G)	FAT (G)
Chicken + Rice + Broccoli	200 g chicken breast, 200 g rice (cooked), 150 g broccoli, 10 g olive oil	50	65	15
Eggs + Oats + Fruit	4 whole eggs, 80 g oats, 200 g berries, 200 ml milk	35	65	20
Salmon + Sweet Potato + Greens	180 g salmon, 250 g sweet potato, 100 g spinach, 10 g butter	40	55	18
Beef Stir-Fry + Rice	180 g lean beef, 200 g rice (cooked), mixed vegetables, soy sauce	45	65	12
Tofu Veggie Bowl	200 g firm tofu, 150 g quinoa (cooked), 100 g edamame, mixed vegetables	35	55	14

19.3 Hand-Portion Guide (When Tracking Is Not Possible)

Table 19.3 Hand-Portion Reference

PORTION	APPROXIMATE SERVING	PROTEIN/CARBS/FAT
1 palm (protein)	20–30 g protein (chicken, fish, meat, tofu)	Protein
1 cupped hand (carbs)	20–30 g carbs (rice, oats, pasta, potato)	Carbs
1 thumb (fat)	10–15 g fat (oil, butter, nut butter, avocado)	Fat
1 fist (vegetables)	Approximately 1 cup (broccoli, spinach, peppers)	Fibre & micronutrients

CHAPTER SUMMARY

Meal structure should fit the client's lifestyle first, not a prescribed template. Hand portions are a practical alternative to tracking. Four meals optimise protein distribution but 3 meals works if per-meal leucine threshold is met. The best meal plan is the one the client can adhere to.

◈ MAKE THIS WORK FOR YOU

Your meal structure should reflect your daily schedule, not a bodybuilding template. A nurse working 12-hour shifts with no dedicated break time needs a different structure from a remote worker with a flexible schedule. The best meal plan is not the one that is theoretically optimal for MPS distribution — it is the one you can follow consistently without it feeling like a second job.

If you have a predictable daily schedule (e.g., office hours, remote work): Three or four evenly spaced meals work well. Prepare breakfast or lunch the night before to reduce decision fatigue. The key is not the exact timing but ensuring each meal meets the leucine threshold (0.4 g/kg per meal of high-quality protein).

If your schedule is unpredictable (shift work, travel, clinical setting): Keep 1–2 high-protein, portable options available at all times (protein bars, ready-to-drink shakes, pre-prepared chicken and rice containers). Your fallback is to prioritise total daily protein even if distribution is suboptimal. Total daily intake is always the highest-priority variable.

If you prefer 2 larger meals (common with intermittent fasting): Each meal must be more protein-dense to hit the leucine threshold. A 70 kg person needs approximately 50–60 g of protein per meal (0.7–0.8 g/kg per meal). This requires intentional meal design — lean meats, egg whites, or protein isolates — and is harder to achieve with plant-based proteins.

Hand portions as a fallback: If you travel frequently or cannot track, the hand-portion method (1 palm protein, 1 cupped hand carbs, 1 thumb fat, 1 fist vegetables per meal) maintains reasonable macro balance without counting. It is approximately 25–30 g protein, 25–35 g carbs, 10–15 g fat per meal — close enough for maintenance contexts.

20. Hydration Protocol

20.1 Daily Schedule

Table 20.1 Daily Hydration Schedule (3 L Target)

TIME	VOLUME	PURPOSE
Upon waking	500 mL	Replenish overnight losses
Mid-morning	500 mL	Maintain hydration
Lunch	500 mL	Meal accompaniment
Pre-training (2 hrs before)	500–600 mL	Ensure training euhydration
Intra-training (if applicable)	200–300 mL every 15 min	Sweat replacement
Post-training	500–1,000 mL	Rehydrate from losses
Evening	500 mL	Before-cutoff hydration

20.2 Electrolyte Additions by Context

Plain water is sufficient for sessions under 60 minutes at moderate intensity. For longer sessions, warm conditions, or high-sweat individuals, add: 300–600 mg sodium, 30–60 mg magnesium per litre. Commercial electrolyte products should be checked for sugar content (which may be unwanted in fat loss contexts). A simple alternative: 1 g salt (approximately 390 mg sodium) + water.

CHAPTER SUMMARY

Total daily water = 2–5+ L depending on activity. Hydration protocol should be scheduled, not ad hoc. Electrolytes become critical in sessions >60 min or hot environments. Urine colour is the best daily screening tool.

◈ MAKE THIS WORK FOR YOU

The scheduled hydration protocol in this chapter is a template, not a rule. If you find it difficult to drink 500 mL upon waking, start with 250 mL and build up over 2–3 weeks. If you wake up 2–3 times per night to urinate, your evening hydration needs adjustment — stop fluid intake 2 hours before bed or reduce evening volume.

If you struggle to drink enough water: Use environmental cues rather than willpower. Keep a 1 L water bottle on your desk and aim to refill it twice during the workday. Add flavouring (lemon, cucumber, electrolyte tabs) if plain water is unappealing. Set a phone reminder every 90 minutes. The goal is habit, not heroism.

If you train in a hot climate or gym without air conditioning: Your hydration requirements are 30–50% higher than the standard recommendations even at the same activity level. Monitor your sweat rate (weigh before and after training), and add 300–600 mg sodium per litre of fluid during sessions longer than 60 minutes. Dark urine the following morning is a sign you started the next day already dehydrated.

Caffeine and hydration: The diuretic effect of caffeine is mild and transient — regular coffee consumption does not cause chronic dehydration. Your morning coffee counts toward your daily fluid intake. However, consuming caffeine late in the day can disrupt sleep quality independent of its mild diuretic effect.

21. Body Recomposition Protocol

21.1 Eligibility Criteria

Body recomposition (simultaneous fat loss and muscle gain) requires specific conditions: the individual must have a body fat percentage above approximately 15% (male) or 25% (female) AND be a training novice or returning detrained individual. Experienced trainees below these body fat thresholds should expect minimal recomposition and instead pursue targeted surplus or deficit phases.

21.2 Recomp Nutrition Protocol

Recomp Macronutrient Protocol

1. **Calories:** 100–300 kcal below maintenance (small deficit)
2. **Protein:** 2.0–2.4 g/kg/day (elevated to support MPS during deficit)
3. **Fat:** 0.8–1.0 g/kg/day (minimum floor for hormonal health)
4. **Carbohydrate:** Remaining calories; concentrate around training windows
5. **Training:** Progressive resistance training 3–5 days/week, sufficient volume for MPS stimulus
6. **Rate expectation:** 0.3–0.5% body weight loss per week (if any); waist measurement and visual changes more useful than weight

21.3 Rate of Progress and When to Switch

Recomp is slower than dedicated bulking or cutting phases. After 8–12 weeks, assess: if body weight is stable but waist measurement is decreasing and performance is stable or increasing, continue. If no changes in any metric for 4 consecutive weeks, transition to a dedicated surplus or deficit phase. The recomp has a limited lifespan in a given individual; once the hormonal and metabolic conditions that permit it shift (lower body fat, higher training age), it ceases to be effective.

CHAPTER SUMMARY

Body recomposition requires specific conditions — body fat >15% (men) or >25% (women) combined with novice or returning trainee status. Expect slow progress at 0.3–0.5% body weight loss per week if any change occurs. Recomp has a limited lifespan; transition to a dedicated surplus or deficit phase after 8–12 weeks if no progress is observed.

◈ MAKE THIS WORK FOR YOU

Body recomposition is powerful but context-dependent, and the most common mistake is attempting it when you are not eligible. If you have been training consistently for more than 2 years and your body fat is already below 15% (male) or 25% (female), body recomposition will be so slow that it is practically undetectable. In this case, dedicated surplus or deficit phases produce better results in less time.

If you are eligible for recomposition (novice or returning after a break): The protocol is straightforward: a small deficit (100–300 kcal below maintenance), high protein (2.0–2.4 g/kg/day), and progressive resistance training 3–5 days per week. Track progress using waist measurements and progress photos, not just the scale — the scale may stay the same while body composition improves.

If you have been recomping for 8–12 weeks with no visible changes: Transition to a dedicated phase. If you are lean enough, run a 8–12 week surplus (+200–300 kcal/day) to build muscle, followed by a maintenance phase, then a 8–12 week deficit (−300–500 kcal/day) to reveal the muscle. The separation of goals produces faster results than trying to achieve both simultaneously.

If you are returning after a layoff (detrained): You are an ideal candidate for recomposition. The phenomenon is sometimes called "muscle memory" — your body regains muscle and loses fat simultaneously for 4–8 weeks before the conditions shift. Take advantage of this window by training consistently and eating at or slightly below maintenance with high protein.

22. Fat Loss Plateau Decision Tree

22.1 Primary Weight Trend Assessment

Before diagnosing a plateau, confirm it exists. Use a 7–14 day rolling average of daily weigh-ins. A "plateau" is defined as no downward trend in the moving average over 2–3 weeks while maintaining adherence. Week-to-week fluctuations of 1–2 kg are normal and do not constitute a plateau.

22.2 Decision Tree

Fat Loss Plateau Diagnostic Protocol

1. **Adherence audit:** Is the client tracking accurately? (Hidden calories: cooking oils, condiments, mindless eating, "free foods" that are not free). → If no, re-educate on tracking. → If yes, proceed.
2. **Weight trend confirmed?** 3+ weeks with no downward trend in 7-day moving average. → If no, continue current plan with patience. → If yes, proceed.
3. **Metabolic adaptation check:** Has the client lost >10% of starting body weight? Has the deficit been maintained for >12 weeks? → If yes, consider diet break (2–4 weeks at maintenance) or reverse diet.
4. **Strength/recovery cross-check:** Is training performance declining? Is sleep quality reduced? Are there signs of overreaching? → If yes, reduce training volume by 10–20% or add a deload week.
5. **Adjustment:** Reduce calories by 100–200 kcal/day OR increase NEAT (steps, daily movement) by 2,000–3,000 steps/day. Reassess after 2 weeks.

22.3 Population-Specific Triage

- **Female athletes:** Rule out low energy availability (see Chapter 27). Weight stabilisation during the luteal phase is normal and not a plateau.
- **PCOS:** Reduce carbohydrate load; increase fibre; consider inositol supplementation (4 g/day myo-inositol + 400 mcg folic acid).
- **Older adults (50+):** Protein increases to ≥2.0 g/kg/day; rate expectations reduced by 20–30%.
- **ED history / disordered eating:** Do not reduce calories further. Refer to Gentle Entry Protocol (Chapter 23) or a mental health professional.

CHAPTER SUMMARY

Confirm a plateau with a 7–14 day rolling average before intervening. Rule out tracking drift first — hidden calories from oils, condiments, and mindless eating are the most common cause. Metabolic adaptation occurs after >12 weeks or >10% body weight loss, warranting a diet break of 2–4 weeks at maintenance to restore leptin and thyroid function. Adjust by 100–200 kcal and reassess after 2 weeks.

◈ MAKE THIS WORK FOR YOU

A fat loss plateau almost always has a modifiable cause — the key is identifying which one applies to you before making adjustments. The decision tree is designed to prevent the most common mistake: cutting calories further when the real problem is tracking drift, metabolic adaptation, or NEAT decline.

Step 1 — Audit tracking accuracy (this is the most likely cause): For 3–5 days, weigh and log everything that passes your lips, including cooking oils, condiments, drinks, and "tastes" during food preparation. Most people underestimate their intake by 20–40%. If your actual intake is 400–800 kcal higher than you thought, the plateau is not metabolic — it is mathematical.

Step 2 — Check NEAT: Compare your step count from the first 2 weeks of your diet to your current step count. A reduction from 8,000 to 5,000 steps/day accounts for approximately 150–200 kcal/day — enough to stall fat loss entirely. If this has happened, do not drop calories further. Instead, restore step count to baseline and re-evaluate.

Step 3 — Consider a diet break: If you have been in a deficit for more than 12 weeks or lost more than 10% of your starting body weight, your leptin and thyroid signalling are suppressed. A 2-week diet break at maintenance calories restores these hormones and resets metabolic rate, often making subsequent fat loss easier than continuing to cut calories lower.

Women: Remember that the luteal phase (days 15–28 of a 28-day cycle) causes 1–3 kg of water retention. A "plateau" that coincides with this phase is not a real plateau — continue the current plan and reassess after your next menstrual phase begins.

23. Gentle Entry Protocol for ED Risk

23.1 Principle: Add Never Subtract

The Gentle Entry Protocol is designed for clients with eating disorder history, ED screening flags, or high fixation risk. The programme avoids any form of restriction, tracking, or deficit. The core principle is addition, not subtraction: instead of removing foods, we add foods that improve overall nutritional quality.

23.2 Three Levels of Entry

Table 23.1 Gentle Entry Levels

LEVEL	APPROACH	DURATION
Level 1: Awareness Only	No dietary changes. Daily protein target awareness only. "Just notice what you eat."	2-4 weeks
Level 2: Gentle Structure	Add 1 serving protein to existing meals. Add 1 serving vegetables. No removals, no tracking.	4-8 weeks
Level 3: Full Protocol Entry	Introduce hand-portion guidance only (no tracking numbers). Structured meal times without restriction.	As tolerated

23.3 The 10-Minute Rule for Binge Urges

When a binge urge arises, wait 10 minutes before acting. Use the time to identify the trigger (emotional, restrictive, environmental). After 10 minutes, if the urge persists, eat a planned portion. This creates a boundary between impulse and action without creating restriction guilt.

23.4 Professional Integration

Any client with active ED symptoms, diagnosed eating disorder, or significant ED screening flags should be working with a registered dietitian or therapist alongside the coaching programme. The coach's role is supportive and in-context, not therapeutic.

CHAPTER SUMMARY

"Add never subtract" is the guiding principle for ED-risk clients. Level 1 (awareness only) should be maintained for 2–4 weeks before any dietary changes are introduced. The 10-minute rule creates a boundary between impulse and action without creating restriction guilt. The coach's role is supportive, not therapeutic — refer to a dietitian or therapist when active ED flags are present.

◈ MAKE THIS WORK FOR YOU

If you or your client has a history of disordered eating, traditional nutrition coaching can do more harm than good. The Gentle Entry Protocol exists precisely to prevent this: it removes the pressure of tracking, restriction, and numbers-based progress, replacing them with additive, intuitive changes that improve nutrition quality without triggering fixation.

If you are self-coaching with a history of ED: Do not track calories or macros. Do not weigh yourself daily. Use Level 1 (Awareness Only) for 2–4 weeks minimum — simply notice what you eat without judgment. Progress in this phase is measured by your relationship with food, not by changes in body composition. If you feel more anxious about food during this phase, slow down.

The 10-minute rule: This is a practical tool for binge urge management. When the urge arises, set a timer for 10 minutes. Do not try to suppress the urge — just delay the response. Use the time to identify the trigger (was I overly restrictive today? is this emotional? am I actually hungry?). After 10 minutes, if the urge persists, eat a planned portion. The goal is not elimination of the behaviour but creating space between impulse and action.

When to seek professional help: If you experience any of the following, refer to a registered dietitian or therapist specialising in eating disorders: regular binge episodes (>1 per week), purging behaviours, extreme dietary restriction (<1,200 kcal/day without medical supervision), significant weight changes not related to a structured health plan, or obsessive thoughts about food that interfere with daily functioning. The coach's role is supportive, not therapeutic.

24. Intermittent Fasting Decision Protocol

24.1 When IF Works and When It Fails

Intermittent fasting (IF) is a meal-timing strategy, not a calorie-control strategy. It works when it improves adherence by reducing decision fatigue or naturally limiting the eating window. It fails when the compressed feeding window leads to overeating (due to extreme hunger), under-recovery (due to insufficient per-meal protein), or energy crashes.

24.2 Decision Tree

IF Eligibility Assessment

1. **Contraindications:** Past or present eating disorder, female athlete with menstrual disruption, Type 1 diabetes, pregnancy/lactation, underweight (BMI < 18.5). → If any, do not recommend IF.
2. **GOAL:** Is the goal weight loss? IF is a tool for adherence, not a direct metabolic advantage. If the client expects IF to "boost metabolism," educate and redirect.
3. **Recovery check:** Can the client consume ≥0.4 g/kg protein within the first post-fasting meal? Can training performance be maintained?
4. **Trial protocol:** 14:10 (14-hour fast, 10-hour eating window) for 2 weeks, then assess: sleep, hunger, training performance, adherence. If positive, proceed. If negative, switch to 12:12 or abandon.

24.3 Fasted Training Considerations

Training in a fasted state shifts substrate utilisation toward fat oxidation (reduced malonyl-CoA inhibition of CPT1). However, this does not produce superior body composition outcomes over fed training when total daily intake is equated. Fasted training may impair performance in high-intensity or high-volume sessions. Practical compromise: fasted low-intensity cardio or accessory work; fed high-intensity main lifts.

CHAPTER SUMMARY

IF is an adherence tool, not a metabolic advantage. Contraindications include ED history, female athletes with menstrual disruption, and underweight individuals. Start with a 14:10 protocol and assess after 2 weeks. Fasted training does not produce superior body composition outcomes when total daily intake is equated.

◈ MAKE THIS WORK FOR YOU

Intermittent fasting is a meal-timing strategy that works for some people and fails for others. The distinction depends on how your individual psychology and physiology respond to the restricted eating window, not on any inherent metabolic advantage of fasting itself. If IF makes you feel more in control of your eating and reduces decision fatigue, it may be beneficial. If it makes you obsess about food during the fast or binge during the feeding window, it is counterproductive regardless of the theoretical benefits.

The most reliable test: Try a 14:10 protocol (14-hour fast, 10-hour eating window) for 2 weeks. Track your sleep quality, training performance, hunger levels, and overall mood. If all four are at least as good as your baseline, IF may be a sustainable approach for you. If any one of them deteriorates, the cost outweighs the benefit.

Females: Be particularly cautious with IF. The female reproductive system is sensitive to energy availability signalling, and prolonged fasting windows can disrupt menstrual function even when total daily intake is adequate. If you notice changes in cycle regularity, sleep quality, or temperature regulation, abandon IF and return to a normal eating schedule. This is not a failure of willpower — it is your biology protecting reproductive function.

Fasted training: If you train in a fasted state, keep the session intensity moderate (low-to-medium RPE) and limit duration to under 60 minutes. High-intensity or high-volume training requires carbohydrate availability to maintain performance. There is no body composition advantage to training fasted — the increased fat oxidation during exercise is compensated for by reduced post-exercise fat oxidation when total daily intake is equated.

25. Plant-Based Nutrition Guide

25.1 Protein Quality and Quantity

Plant proteins have lower leucine content and reduced digestibility (PDCAAS scores: soy 0.92–1.0, pea 0.69–0.89, rice 0.50–0.60, wheat 0.42–0.50). To compensate, increase total protein intake by 15–25% beyond omnivore recommendations. Complementary protein pairing (e.g., rice + beans, pea + rice) ensures complete amino acid profiles across the day, though the "complementarity within meals" dogma has been relaxed; total daily amino acid profile is the relevant metric.

25.2 Critical Micronutrients for Plant-Based Athletes

Table 25.1 Plant-Based Athlete Supplement Considerations

NUTRIENT	RISK LEVEL	RECOMMENDATION
Vitamin B12	Critical — deficiency inevitable without supplementation	≥50 mcg/day (cyanocobalamin) or 2,000 mcg once/week
Iron	High — non-haem iron absorption 5-15% vs. haem iron 25%	Pair with vitamin C; avoid tea/coffee within 1 hour; test ferritin annually
Zinc	Moderate — phytates reduce absorption by 30-50%	Soak/sprout legumes and grains; supplement if deficient (15-30 mg/day)
Calcium	Variable — depends on fortified food intake	Fortified plant milks (300-450 mg per serving), calcium-set tofu
Vitamin D	Moderate — reduced sun exposure, vegan D2 vs. D3	1,000-2,000 IU/day; choose plant-based D3 (lichen-derived)
Omega-3 DHA/EPA	Moderate — ALA to DHA conversion <5%	Algal DHA/EPA supplement (200-500 mg/day)
Iodine	Moderate — seaweed is unreliable source	Iodised salt or 150 mcg/day supplement

CHAPTER SUMMARY

Plant-based athletes need 15–25% more total protein to compensate for lower leucine content and reduced digestibility. B12 supplementation is mandatory for vegans. Iron and zinc require attention due to reduced bioavailability from phytates. Pairing complementary proteins across the day is sufficient — within-meal complementarity is not required.

◈ **MAKE THIS WORK FOR YOU**

Adopting a plant-based diet while pursuing body composition goals requires more intentional planning than an omnivorous diet, but it is entirely achievable. The key is to anticipate the nutritional gaps and address them before they become problems, rather than waiting for deficiency symptoms to appear.

Protein structuring for plant-based athletes: Aim for 1.8–2.6 g/kg/day (the higher end of the range). Include at least one serving of soy (tofu, tempeh, edamame) per day for its complete amino acid profile. Use protein blends (pea + rice) for post-training shakes to ensure rapid leucine delivery. The idea that plant proteins are "incomplete" is outdated — what matters is the total amino acid profile across the day, not within a single meal.

Critical supplements checklist for vegans: (1) B12 — non-negotiable, ≥50 mcg/day cyanocobalamin or 2,000 mcg once/week. (2) Vitamin D — 1,000–2,000 IU/day, choose vegan D3 from lichen. (3) Algal DHA/EPA — 200–500 mg/day. (4) Iron — only after testing ferritin, but awareness is key. (5) Zinc — consider 15–30 mg/day if intake from food is low. (6) Iodine — iodised salt or 150 mcg/day supplement.

Practical iron absorption: Plant iron (non-haem) absorption is only 5–15%, compared to 25% for haem iron from animal sources. Pair every iron-rich plant meal with vitamin C (citrus, bell peppers, tomatoes) and separate it from tea, coffee, and calcium-rich foods by at least one hour. Soaking and sprouting legumes and grains reduces phytate content and improves mineral absorption.

26. Complete Nutrition Assessment

A structured nutrition assessment should be conducted at programme intake and reassessed after each phase (typically every 4–6 weeks). The assessment covers seven domains:

26.1 Eating Patterns

- Number of meals per day; typical timing; consistency from day to day
- Eating window (first to last calorie); largest meal of the day
- Training-nutrition interaction: pre/intra/post training eating habits
- Night eating: frequency of eating within 2 hours of sleep

26.2 Protein Assessment

Table 26.1 Protein Gap Identification

QUESTION	ASSESSMENT
Total daily protein (estimate in g/kg)?	Compare to target from Table 6.1
Protein per meal (estimate in g)?	<20 g signals insufficient leucine threshold
Primary protein sources?	Quality and leucine content assessment
Post-training protein timing?	Within 2-4 hours of training
Pre-sleep protein consumed?	30-50 g casein or equivalent recommended

26.3 Vegetable & Fibre Assessment

- Current daily vegetable servings (target: 3–5 servings)
- Current daily fibre intake (target: 25–35 g/day for men, 21–28 g/day for women in deficit)
- Variety: variety of colourful vegetables across the week (different polyphenol profiles)
- Fermented food intake (target: 1–3 servings/week for microbiome diversity)

26.4 Hydration Assessment

- Daily water intake (litres); compare to Table 13.1 targets
- Urine colour throughout the day (pale straw = good)
- Training hydration habits (pre/intra/post)

- Caffeine intake (diuretic? tolerance-adjusted? cycle pattern?)
- Alcohol intake (frequency, volume, timing relative to training)

26.5 Eating Behaviour Screening

- Relationship with food: guilt, anxiety, or distress around eating
- Dietary restraint: rigid rules vs. flexible boundaries
- Binge eating: frequency, triggers, compensatory behaviours
- Food tracking attitude: tool or obsession?
- Body image: realistic goals vs. dysmorphic concern

26.6 Supplement Use Audit

- Current supplement list (with doses and brands)
- Third-party testing status (Informed Sport, NSF)
- Perceived benefits vs. actual effects
- Budget allocation for supplements
- Stack interaction check

26.7 Nutrition Priority Matrix

Based on the assessment, prioritise interventions in order of impact:

1. **Total protein** — bring to 1.6–2.2 g/kg/day (first priority, highest impact)
2. **Protein distribution** — even across 3–5 meals (second priority)
3. **Calorie targeting** — establish appropriate deficit/surplus (third priority)
4. **Fibre & vegetables** — food quality, satiety, micronutrients (fourth priority)
5. **Hydration** — daily and training-specific (fifth priority)
6. **Nutrient timing** — training windows, pre-sleep (sixth priority)
7. **Supplements** — targeted, evidence-based additions (last priority, after food is optimised)

CHAPTER SUMMARY

The 7-domain assessment covers eating patterns, protein, fibre, hydration, eating behaviour, supplements, and a priority matrix. Always prioritise protein first — it has the highest impact on body composition outcomes. The assessment should be repeated every 4–6 weeks.

◈ MAKE THIS WORK FOR YOU

The Nutrition Assessment is designed to be used as a self-reflection tool, not just a coach-facing diagnostic. Every 4–6 weeks, run yourself through the 7 domains and identify the single domain that would produce the biggest improvement if addressed. The priority matrix exists to prevent you from chasing low-impact changes (perfecting nutrient timing) while neglecting high-impact fundamentals (increasing total protein).

Running your self-assessment: Start with Domain 1 (Eating Patterns) — compare your current meal schedule, consistency, and training-nutrition alignment to the ideal state. Then move through the remaining domains one by one. For each domain, identify one small change that would improve your score. The goal is not to fix all 7 domains at once — it is to make one sustainable improvement per assessment cycle.

The priority matrix in practice: If your total protein is below 1.6 g/kg/day, that is the only thing that matters until it is fixed. If your protein is adequate but your fibre intake is below 25 g/day, fibre becomes the priority. If both are adequate, then consider carbohydrate periodisation or supplement optimisation. The order of operations prevents you from getting distracted by marginal gains while the fundamentals are still suboptimal.

Eating behaviour screening (Domain 5): This is the most overlooked domain. If your relationship with food is characterised by guilt, anxiety, or rigidity, no amount of macronutrient optimisation will produce sustainable results. Address the psychology of eating before you attempt advanced nutrition protocols. The Gentle Entry Protocol (Chapter 23) may be more appropriate than a standard deficit phase.

COACHING APPLICATION

Fat Loss Plateau in an Intermediate Client

Scenario: A 40-year-old male client, 92 kg, 22% body fat, has been following a 2,200 kcal protocol for 10 weeks. He lost 6 kg in weeks 1-6 but has seen no weight trend change in the past 4 weeks. He trains 4x/week (resistance), steps have dropped from 8,000 to 5,000/day due to work stress, and sleep has declined to 6 hours/night.

Assessment: (1) Confirmed plateau — 4 weeks with no downward trend in 7-day moving average. (2) Tracking audit reveals no hidden calories. (3) >10% body weight lost (6.5%) approaching the metabolic adaptation threshold. (4) NEAT has dropped 40% — this is the most likely driver of the plateau. (5) Reduced sleep is elevating cortisol, further impairing fat oxidation and recovery.

Intervention: (1) Reverse diet to 2,500 kcal for 2-week diet break to restore leptin and thyroid signalling. (2) Target 8,000+ steps/day as a non-negotiable NEAT baseline. (3) Sleep hygiene protocol: no screens 60 min before bed, magnesium glycinate 200 mg, target 7.5+ hours. (4) Maintain protein at 2.0 g/kg (184 g/day). (5) After 2-week diet break, resume deficit at 2,300 kcal. (6) Reassess after 3 weeks on the new deficit.

Key Takeaway: Plateaus are rarely about needing a larger deficit. Metabolic adaptation, NEAT drift, and sleep quality must be assessed first. A diet break often restores the conditions for fat loss more effectively than further calorie reduction.

Domain 4: Special Populations & Coaching

27. Female Physiology & Nutrition

27.1 The Menstrual Cycle and Energy Metabolism

The menstrual cycle creates predictable physiological fluctuations that affect nutrition and training. Oestrogen and progesterone modulate substrate metabolism, appetite, hydration status, and thermoregulation across the cycle. Understanding these changes prevents misinterpreting normal cycle-driven fluctuations as stalls or regressions.

Table 27.1 Nutritional Considerations Across the Menstrual Cycle

PHASE	DAYS (28-DAY CYCLE)	METABOLIC CHANGES	NUTRITIONAL ADJUSTMENTS

Early Follicular (menses)	1-5	Low oestrogen, low progesterone; increased insulin sensitivity	Standard carbohydrate utilisation; potentially lower appetite
Late Follicular	6-14	Rising oestrogen; improved insulin sensitivity; peak muscle recovery	Optimal window for high training volumes; standard nutrition
Luteal	15-28	High progesterone; reduced insulin sensitivity; increased REE by 5-10%; increased sodium/water retention (1-3 kg apparent weight gain); increased core temperature	May benefit from slightly higher carbohydrate intake; increase magnesium and iron; expect 1-3 kg scale increase (water, not fat); increase fluid intake; prioritise sleep

27.2 Fat Loss Rate Adjustments for Females

Expected fat loss rates are lower for females than males due to lower resting energy expenditure relative to body size, hormonal differences in fat mobilisation (higher alpha-2 adrenergic receptor density in gluteofemoral adipose tissue), and the metabolic cost of the menstrual cycle. Recommended rates: 0.5–0.7% of body weight per week (vs. 0.7–1.0% for males). Scale weight should never be interpreted during the luteal phase without accounting for water retention.

27.3 Low Energy Availability (LEA) and REDs

Female athletes are at elevated risk of Relative Energy Deficiency in Sport (REDs). LEA occurs when energy intake is insufficient to support both training demands and normal physiological functions. Consequences include menstrual disruption (functional hypothalamic amenorrhea), bone mineral density loss, impaired immune function, cardiovascular dysfunction, and psychological impairment. **Protocol:** Increase intake by 200–500 kcal/day; reduce exercise energy expenditure if necessary; aim for EA >45 kcal/kg FFM/day. Return of menses may take 3–12+ months even after energy restoration.

27.4 PCOS-Specific Nutrition

Polycystic ovary syndrome is characterised by insulin resistance, hyperandrogenism, and anovulation. Nutrition approach: reduce glycaemic load (not necessarily total carbohydrate), increase fibre (30–40 g/day), emphasise monounsaturated fats, and ensure adequate protein. Supplement considerations: myo-inositol (4 g/day) + D-chiro-inositol (400 mcg/day) in physiologic ratio 40:1; vitamin D (if deficient); omega-3s (1.5–3 g EPA+DHA/day). Weight loss of 5–10% can restore ovulation in many cases.

CHAPTER SUMMARY

Female clients require adjusted fat loss expectations (0.5-0.7%/week vs 0.7-1.0% for males). The luteal phase causes 1-3 kg water retention — do not interpret scale weight during this phase. Low energy availability (LEA) is the most critical condition to screen for. PCOS clients benefit from lower glycaemic load and higher fibre. Menstrual cycle awareness helps distinguish normal fluctuations from true plateaus.

◈ MAKE THIS WORK FOR YOU

Female physiology is not a deviation from a male norm — it is a distinct metabolic reality that requires its own nutritional framework. If you are a female athlete or coach a female client, the most important shift is abandoning the expectation that the same protocols that work for males will work equally well for females. They will not — not because females are "less compliant" or "less disciplined," but because oestrogen and progesterone fundamentally alter substrate metabolism, fluid balance, and energy regulation across the menstrual cycle.

Tracking your cycle for better nutrition decisions: Log your cycle phase alongside your weight, training performance, and subjective energy for 2–3 months. You will likely observe patterns: weight increases of 1–3 kg during the luteal phase (water retention, not fat gain), reduced high-intensity performance in the late luteal phase, increased appetite and carbohydrate cravings in the week before menstruation. These are not problems to be solved — they are normal physiology to be accommodated.

If you have PCOS: Your nutritional priority is managing insulin resistance, not counting calories with extreme precision. A lower glycaemic load approach (reducing refined carbohydrates, increasing fibre to 30–40 g/day) is often more effective than a standard calorie deficit. Supplementing with myo-inositol (4 g/day) and maintaining adequate vitamin D status are evidence-supported adjuncts. A 5–10% reduction in body weight can restore ovulation in many cases.

Low energy availability screening: LEA is the most dangerous nutrition condition for female athletes because it is often invisible until significant harm has occurred. If you have lost your menstrual cycle, experience sleep disruption, have frequent injuries or illness, or feel cold when others are comfortable, your energy intake may be too low relative to your expenditure. The protocol is not to push harder — it is to eat more and train less, which is psychologically difficult but physiologically necessary.

28. Gut Health, Digestion & the Microbiome

28.1 The Microbiome and Nutrient Partitioning

The gut microbiome influences body composition through multiple mechanisms: energy harvest (certain bacterial phyla extract more calories from indigestible fibre), nutrient partitioning (the Firmicutes:Bacteroidetes ratio is elevated in obesity), inflammation (LPS from gram-negative bacteria triggers systemic inflammation and insulin resistance), and appetite regulation (gut-derived peptides: GLP-1, PYY, ghrelin).

28.2 Fibre: The Three Types

Table 28.1 Fibre Types and Their Roles

TYPE	MECHANISM	SOURCES	DAILY TARGET
Soluble fibre	Forms gel; slows gastric emptying; binds bile acids; lowers cholesterol; improves glycaemic control	Oats, barley, psyllium, apples, carrots, beans	10–15 g/day
Insoluble fibre	Increases stool bulk; promotes bowel regularity; reduces transit time	Wheat bran, nuts, vegetables, whole grains	10–15 g/day
Prebiotic fibre	Fermentable by gut bacteria; produces short-chain fatty acids (SCFAs: butyrate, propionate, acetate); feeds beneficial bacteria	Garlic, onions, leeks, asparagus, bananas, oats, chicory root (inulin)	5–10 g/day (introduce gradually to avoid bloating)

28.3 Fibre Progression Protocol

Introduce fibre gradually to minimise bloating and digestive discomfort: **Level 1** (Week 1–2): target 15 g/day (add 1–2 servings of vegetables or fruit). **Level 2** (Week 3–4): target 20–25 g/day (add legumes or whole grains). **Level 3** (Week 5+): target 25–35 g/day (full diversity). Water intake must increase proportionally (approximately 1 L per 10 g fibre) to prevent constipation. Troubleshooting: persistent bloating suggests insufficient water, too-rapid progression, or a specific FODMAP sensitivity.

28.4 Fermented Foods

Regular consumption of fermented foods (kefir, yogurt, sauerkraut, kimchi, kombucha) increases gut microbiome diversity and reduces inflammatory markers. Target: 1–3 servings per week minimum. The diversity of bacterial strains matters more than the quantity of any single food.

CHAPTER SUMMARY

The microbiome influences body composition through energy harvest, inflammation, and appetite regulation. Fibre intake should progress slowly (15→25→35 g/day over 5+ weeks) to minimise digestive discomfort. All three fibre types (soluble, insoluble, prebiotic) are necessary. Fermented foods (1-3 servings/week) improve microbiome diversity. Water must increase proportionally with fibre.

◈ MAKE THIS WORK FOR YOU

Gut health is deeply individual — what works for one person triggers bloating in another. The general principles in this chapter (prioritise fibre, eat fermented foods, stay hydrated) are universal, but the implementation must be tailored to your tolerance and baseline. A client who has been eating 10 g of fibre per day cannot jump to 35 g in one week without significant digestive distress.

The fibre progression in practice: Start at your current intake and add 5 g of fibre per week for 4–6 weeks until you reach 25–35 g/day. Each time you add fibre, increase your water intake proportionally (approximately 1 L per 10 g of fibre). If you experience persistent bloating at a given level, maintain that level for an additional 1–2 weeks before progressing, or investigate whether a specific FODMAP group (fermentable carbohydrates found in wheat, onions, garlic, beans, and some fruits) is the trigger.

If you have a diagnosed digestive condition (IBS, IBD, SIBO): The generic high-fibre recommendation may not be appropriate for you. A low-FODMAP elimination phase followed by systematic reintroduction may be necessary to identify your trigger foods. Work with a dietitian who specialises in digestive health — fibre is beneficial, but the wrong type or too-rapid introduction can exacerbate symptoms.

Fermented foods: Start with one serving (100–200 g of yogurt, kefir, sauerkraut, or kimchi) per day, not per week. The diversity of bacterial strains is more important than the volume of any single food. If you do not tolerate dairy-based ferments, try vegetable ferments (sauerkraut, kimchi) or water kefir.

29. Protein Sources

Table 29.1 Common Protein Sources (per 100 g as prepared)

FOOD	CALORIES	PROTEIN (G)	CARBS (G)	FAT (G)	NOTES
Chicken breast (skinless, grilled)	165	31	0	3.6	Leanest common protein; versatile
Lean beef (sirloin, grilled)	206	26	0	10.6	Rich in zinc, iron, B12; choose 90%+ lean
Salmon (Atlantic, wild)	208	20	0	13.4	Omega-3 EPA+DHA, vitamin D
Tuna (canned in water)	116	25	0	0.8	Convenient; limit to 2-3 servings/week (mercury)
Whole eggs	155	13	1.1	11	Complete protein; choline; yolk contains most micronutrients
Egg whites	52	11	0.7	0.2	Pure protein; low calorie
Greek yogurt (plain, nonfat)	59	10	3.6	0.4	High protein:density ratio; casein-dominant
Cottage cheese (1%)	72	11	3.0	1.0	Casein-rich; excellent pre-sleep option
Whey protein isolate	400	90	4.0	1.5	Per 100 g powder; ~25 g protein per 28 g scoop
Casein (micellar)	380	85	5.0	1.0	Sustained release 4–6 h; pre-sleep
Tofu (firm, calcium-set)	78	8	2.0	4.5	Complete protein; calcium-rich; versatile
Tempeh	195	19	9.0	11	Fermented soy; higher protein than tofu
Seitan	150	24	8.0	2.0	Wheat gluten; high protein, low fat; incomplete alone
Lentils (cooked)	116	9	20	0.4	Protein + fibre + iron; pair with grains
Chickpeas (cooked)	139	7.6	27	2.6	Protein + fibre; versatile in meals

CHAPTER SUMMARY

Protein sources table reference. Prioritise variety across animal and plant sources. Leucine density per serving is the key quality metric for muscle protein synthesis.

◈ MAKE THIS WORK FOR YOU

Use this table to design meals that hit your protein targets efficiently. If your goal is 170 g of protein per day across 4 meals, you need approximately 40–45 g per meal. That means each meal should contain roughly 130–150 g of chicken breast, 160–180 g of lean beef, or a combination of sources (e.g., 100 g chicken + 2 eggs + 150 g Greek yogurt).

Practical meal building: Choose 2–3 protein sources you enjoy and can prepare consistently. The best protein source is the one you will eat regularly, not the one with the highest leucine score. If you are plant-based, use higher quantities and combine sources (e.g., tofu + lentils in the same meal) to ensure amino acid adequacy.

Budget-friendly protein: Whole eggs, Greek yogurt, cottage cheese, canned tuna, and whey protein concentrate offer the best protein-to-cost ratio. Chicken breast and lean beef offer the best protein-to-calorie ratio for fat loss phases.

30. Carbohydrate Sources

Table 30.1 Common Carbohydrate Sources (per 100 g as prepared)

FOOD	CALORIES	CARBS (G)	PROTEIN (G)	FAT (G)	FIBRE (G)
White rice (cooked)	130	28	2.7	0.3	0.4
Brown rice (cooked)	123	26	2.7	1.0	1.6
Oats (rolled, cooked)	71	12	2.5	1.4	1.7
Potato (white, baked, flesh)	93	21	2.5	0.1	2.2
Sweet potato (baked, flesh)	90	21	2.0	0.1	3.3
Whole wheat pasta (cooked)	131	25	5.0	0.5	3.0
Quinoa (cooked)	120	21	4.4	1.9	2.8
Banana (medium, 118 g)	105	27	1.3	0.4	3.1
Whole wheat bread (1 slice, ~30 g)	80	14	3.5	1.0	2.0

CHAPTER SUMMARY

Carbohydrate sources table reference. Choose based on training context — higher GI around training, lower GI at non-training meals. Prioritise whole-food carbohydrate sources for micronutrient density.

◈ MAKE THIS WORK FOR YOU

Your carbohydrate source selection should match your training context and digestive tolerance. Around training, choose faster-digesting sources (white rice, potatoes, bananas, white bread) to minimise digestive discomfort and provide rapid glucose availability. At non-training meals, choose slower-digesting, higher-fibre sources (oats, sweet potato, quinoa, legumes) to improve satiety and glycaemic control.

If you have digestive sensitivity to certain carbs (bloating from beans, heaviness from oats): Choose alternatives that work for your gut. White rice is well-tolerated by almost everyone. Potatoes (without excessive fat) are also well-tolerated and provide substantial carbohydrate with minimal gut burden. There is no rule that carbohydrate sources must be "healthy" — what matters is that they fit your calorie and micronutrient targets.

If you are in a fat loss phase: Prioritise vegetables and higher-fibre carbohydrate sources to improve satiety on a lower calorie budget. A 200 g serving of broccoli provides only 70 kcal and significant fibre and micronutrients, whereas 200 g of white rice provides approximately 260 kcal with minimal fibre. The choice is not about "good" vs "bad" carbs — it is about fitting the right carb density to your current goal.

31. Fat Sources

Table 31.1 Common Fat Sources

FOOD	SERVING	CALORIES	TOTAL FAT (G)	MUFA (G)	PUFA (G)	SFA (G)
Olive oil (extra virgin)	1 tbsp (15 mL)	119	13.5	10.0	1.5	1.9
Almonds	30 g	170	15	9.5	3.5	1.1
Avocado	100 g (half)	160	15	10.0	1.8	2.1
Peanut butter (natural)	2 tbsp (32 g)	190	16	7.5	4.5	3.3
Mixed nuts	30 g	180	16	7.5	5.5	1.7
Butter	1 tbsp (14 g)	102	11.5	3.3	0.4	7.2
Flaxseed (ground)	1 tbsp (7 g)	37	3.0	0.6	2.2 (ALA)	0.3

CHAPTER SUMMARY

Fat sources table reference. Prioritise MUFA and PUFA sources, including omega-3-rich foods regularly. SFA from whole food sources is acceptable within <10% of total calories.

◈ MAKE THIS WORK FOR YOU

Your fat source selection should prioritise unsaturated fats for their anti-inflammatory and cardioprotective effects. Extra virgin olive oil, avocado, nuts, and seeds provide MUFA and PUFA alongside beneficial phytochemicals. Fatty fish (salmon, mackerel, sardines) provide the added benefit of pre-formed EPA/DHA, which is difficult to obtain from plant sources alone.

Practical fat distribution: Aim to include at least one source of unsaturated fat in each meal. Drizzle olive oil over roasted vegetables, add half an avocado to your lunch, or include a handful of almonds with breakfast. This not only improves your fatty acid profile but also enhances the absorption of fat-soluble vitamins (A, D, E, K) from the vegetables in that meal.

If you are on a tight calorie budget: You do not need to add extra fats to your meals if the fat from your protein sources (eggs, meat, fish) meets your minimum target. What matters is that you do not drop below the fat floor and that at least some of your fat comes from unsaturated sources. A serving of olives, a tablespoon of flaxseed, or a small portion of nuts can provide the necessary fatty acid diversity without a significant calorie impact.

32. Vegetables & Fruits

Table 32.1 Vegetable and Fruit Nutrient Density

FOOD	SERVING	CALORIES	FIBRE (G)	KEY MICRONUTRIENTS
Broccoli	100 g	34	2.6	Vit C, K, folate; sulforaphane (Nrf2 activator)
Spinach	100 g	23	2.2	Iron, magnesium, vit K, vit A; oxalates moderate
Bell peppers (red)	100 g	31	2.1	Vit C (190% RDA), vit A, B6
Tomatoes	100 g	18	1.2	Lycopene, vit C, potassium; cooked = more bioavailable lycopene
Carrots	100 g	41	2.8	Beta-carotene (vit A precursor), vit K, potassium
Blueberries	100 g	57	2.4	Anthocyanins, vit C, vit K; antioxidant
Banana	100 g	89	2.6	Potassium, vit B6, vit C; carb-dense fruit
Apple	100 g (medium)	52	2.4	Vit C, quercetin; pectin (soluble fibre)

CHAPTER SUMMARY

Vegetable and fruit reference table. Target 3–5 servings of vegetables and 2–3 servings of fruit daily. Variety across colours ensures a diverse polyphenol profile and broad micronutrient coverage.

◈ MAKE THIS WORK FOR YOU

Vegetables and fruits are the most underutilised tool in body composition nutrition. They provide fibre for satiety, micronutrients for metabolic function, and polyphenols for antioxidant support — all at a low calorie cost. In a fat loss phase, vegetables are your primary tool for maintaining meal volume and satisfaction without exceeding your calorie target.

Practical inclusion: Aim for a "colour rainbow" across the week, not per meal. Different colours represent different polyphenol profiles — red (lycopene from tomatoes, watermelon), orange (beta-carotene from carrots, sweet potato), green (chlorophyll, folate from spinach, broccoli), purple (anthocyanins from blueberries, red cabbage), and white (quercetin from onions, allicin from garlic). Each phytochemical group supports different aspects of health and recovery.

If you struggle to eat enough vegetables: Start each lunch and dinner with a serving of vegetables (either as a first course or mixed into the main dish). Include vegetables in your breakfast (spinach in eggs, bell peppers in omelettes). Use frozen vegetables as a convenient, cost-effective option — they are nutritionally equivalent to fresh and reduce preparation time.

Fruit intake: Do not fear fruit due to its sugar content. The fibre, water content, and polyphenol density of whole fruit make it a net positive for body composition, not a negative. The exception is dried fruit, which is calorie-dense and easy to overconsume. Stick to whole fruit (berries, apples, citrus, melon) for the best nutrient-to-calorie ratio.

33. Complete Nutrition Reference Tables

33.1 Mineral Reference Table (Athlete RDA)

Table 33.1 Key Minerals for Athletic Performance

MINERAL	ATHLETE RDA	TOP FOOD SOURCES	SUPPLEMENT IF...
Sodium	3-5 g/day	Table salt, pickled foods, sports drinks, bone broth	Heavy sweater, training >2 hrs in heat
Potassium	3.5-5 g/day	Banana, potato, spinach, avocado, coconut water, beans	Low fruit/vegetable intake, diuretic use
Magnesium	400-600 mg/day	Pumpkin seeds, almonds, spinach, dark chocolate, beans	Cramps, poor sleep, high stress, deficit phase
Calcium	1,000-1,300 mg/day	Dairy, fortified plant milk, calcium-set tofu, sardines, kale	Low dairy intake, female athlete, <600 mg/day from food
Zinc	11-15 mg/day	Oysters, beef, pumpkin seeds, lentils, cashews	Low red meat intake, vegetarian/vegan, high training volume
Iron	8-18 mg/day	Red meat, spinach, lentils, fortified cereals, liver	Only after testing ferritin; female athletes at higher risk
Iodine	150 mcg/day	Iodised salt, seaweed, fish, dairy	Vegan, no iodised salt, no sea vegetables
Selenium	55-70 mcg/day	Brazil nut (1 nut = ~100 mcg), tuna, sardines, eggs	Vegan in low-selenium region

33.2 Vitamin Reference Table (Athlete RDA)

Table 33.2 Key Vitamins for Athletic Performance

VITAMIN	ATHLETE RDA	TOP FOOD SOURCES	SUPPLEMENT IF...
Vitamin D	1,000-2,000 IU/day	Salmon, UV mushrooms, fortified milk, egg yolks	Low sun exposure, winter, deficiency likely
B12	2.5-5 mcg/day	Meat, fish, dairy, eggs; NOT in plants	Vegan or vegetarian mandatory

Folate (B9)	400-800 mcg DFE	Leafy greens, legumes, asparagus, fortified grains	Pregnancy, low vegetable intake, MTHFR variant
Vitamin C	100-200 mg/day	Citrus, bell peppers, kiwi, broccoli, strawberries	Low fruit/vegetable intake; avoid megadoses near training
Vitamin K2	100-200 mcg/day	Natto, liver, cheese, egg yolks, sauerkraut	With vitamin D supplementation (synergistic)

CHAPTER SUMMARY

Complete reference tables for minerals and vitamins. Use as a quick-reference guide when building meal plans and assessing client intake patterns. Refer to the athlete-specific RDAs when evaluating supplementation decisions.

◈ MAKE THIS WORK FOR YOU

These reference tables are designed for practical use, not academic study. When assessing your own nutrition, cross-reference your typical daily intake against these tables to identify potential gaps. If you rarely consume dairy, your calcium intake likely needs attention. If you follow a vegan diet, B12, iron, zinc, iodine, and omega-3 status all require proactive management. If you train at a high volume, your magnesium and sodium requirements are elevated.

How to use the tables: (1) Identify the nutrients most relevant to your diet type and lifestyle. (2) Estimate your intake from food sources over a typical day. (3) Identify gaps between your intake and the athlete RDA. (4) Address gaps through food choices first, supplementation second. (5) For iron, vitamin D, and B12, consider testing before supplementing to avoid unnecessary or excessive intake.

The supplement interaction table (A.2) is particularly important: Many athletes take multiple supplements without considering how they interact. Zinc and iron compete for absorption. Vitamin D requires K2 and magnesium to function optimally. High-dose zinc depletes copper over time. Review your current supplement stack against this table at least once per training cycle to ensure you are not creating new deficiencies while trying to correct existing ones.

COACHING APPLICATION

Female Client with Suspected LEA

Scenario: A 26-year-old female client, 58 kg, has been training for a physique competition for 16 weeks. She is on 1,600 kcal/day, trains 6x/week (2x/day on 3 days), does 45 min fasted LISS 5x/week. She reports absent menstruation for 3 months, waking at 3 AM unable to fall back asleep, cravings, and irritable mood. Her protein intake is 120 g/day (2.1 g/kg), carbs 140 g, fat 35 g.

Assessment: Classic REDs presentation — menstrual disruption (functional hypothalamic amenorrhea), sleep disturbance, mood changes, and cravings. Fat intake at 0.6 g/kg is at the minimum boundary but may be insufficient given her extreme energy expenditure. Estimated energy availability is approximately 25 kcal/kg FFM/day — well below the 45 kcal/kg FFM/day threshold. Gut health is likely compromised due to chronic low intake and high training volume.

Intervention: (1) Immediate protocol change: increase calories to 1,900 kcal/day (add 300 kcal from carbs and fat). (2) Increase fat to 0.8 g/kg (46 g/day) to support hormonal health. (3) Reduce training load: eliminate fasted cardio, reduce to 5x/week single sessions. (4) Add magnesium glycinate (200 mg before bed) for sleep. (5) Educate on REDs — return of menses may take 3-12 months even after energy restoration. (6) Recommend referral to a sports dietitian and gynaecologist. (7) Reassess energy, sleep, and menstrual status after 4 weeks.

Key Takeaway: Low energy availability is the most serious nutrition condition a coach will encounter in female clients. Performance and body composition goals must be deprioritised until physiological health is restored. Coaches must recognise when their scope has been exceeded and refer appropriately.

Appendix A: Deficiency Screening & Supplement Protocols

A.1 Micronutrient Deficiency Decision Framework

Table A.1 Deficiency Screening and Supplement Decisions

NUTRIENT	RED FLAGS (SYMPTOMS & RISK FACTORS)	TESTING	SUPPLEMENT PROTOCOL	FOOD SOURCES

Vitamin D	Low sun exposure, winter, dark skin, BMI >30, gut malabsorption, muscle weakness, fatigue, frequent illness	Serum 25(OH)D: target 50-80 ng/mL	Deficient: 5,000 IU/day or 50,000 IU/week x 8 wk; Maintenance: 1,000-2,000 IU/day	Salmon, UV mushrooms, fortified foods, egg yolks
Magnesium	Nocturnal cramps, restless legs, anxiety, poor sleep, constipation, high exercise volume, alcohol use, diabetes	RBC magnesium (serum is unreliable)	200-400 mg elemental Mg/day (glycinate for sleep, malate for energy)	Pumpkin seeds, almonds, spinach, dark chocolate, beans
Zinc	Low red meat intake, vegetarian/vegan, frequent infections, poor wound healing, acne, low testosterone symptoms	Serum zinc (fasting)	15-30 mg/day zinc (picolinate or glycinate); add 1-2 mg copper if >15 mg Zn for >3 months	Oysters, beef, pumpkin seeds, lentils, cashews
Iron	Female athlete, vegetarian/vegan, heavy menstruation, fatigue, pallor, exercise intolerance	Ferritin, serum Fe, TIBC, transferrin saturation (essential before supplementing)	Only if low: 18-65 mg elemental Fe/day with vit C; separate from Ca, Zn, tea, coffee	Red meat, liver, spinach, lentils, fortified cereals
B12	Vegan/vegetarian, older adult, metformin use, fatigue, neurological symptoms	Serum B12, homocysteine, MMA	Vegan: ≥50 mcg/day or 2,000 mcg/week; Deficiency: 1,000 mcg/day initially	Only animal foods; fortified foods for vegans
Omega-3 (EPA/DHA)	Low fish intake (<2 servings/week), vegan, high inflammation, joint pain	Omega-3 index (% EPA+DHA in RBC membranes)	1.5-4 g/day EPA+DHA (rTG form preferred)	Fatty fish (salmon, mackerel, sardines); algal oil (vegan)

A.2 Supplement Stack Interactions

Table A.2 Known Supplement Interactions

SUPPLEMENTS	INTERACTION	MANAGEMENT
Zinc + Iron + Calcium	Competitive absorption via DMT1	Separate by ≥4 hours; take Zn with protein, Fe with vit C, Ca with meal
Calcium + Iron	Calcium inhibits non-haem iron absorption	Take iron separate from calcium-rich meals/supplements
Magnesium + Zinc	Synergistic for testosterone; Mg required for Zn absorption	Can be taken together; optimal for evening
Vitamin D + K2 + Mg	K2 directs Ca to bone; Mg activates VDR; synergistic trio	Take together with a meal containing fat
Creatine + Caffeine	Debated interaction; caffeine may blunt creatine uptake acutely	Separate by ≥2 hours or ignore (effect is small and inconsistent)
Omega-3 + Blood thinners	Additive anticoagulant effect at high doses (>3 g/day)	Monitor; inform physician before surgery
Zinc (high dose, >40 mg/day) + Copper	Zinc induces metallothionein, which binds Cu and reduces absorption	Supplement 1–2 mg Cu per 15–30 mg Zn long-term

A.3 Supplement Verification

Only recommend supplements with third-party testing certification. The primary certifying bodies are:

- **Informed Sport** — Gold standard for athletes; batch-tested for banned substances
- **NSF Certified for Sport** — Similar to Informed Sport; widely recognised
- **USP Verified** — Tests for purity, potency, and manufacturing quality (does not test for banned substances)
- **ConsumerLab.com** — Independent testing; subscription required for full access

Appendix B: Nutrient Timing Quick Reference

Table B.1 Nutrient Timing Summary

TIMING	RECOMMENDATION	RATIONALE
Pre-training (2–3 hours before)	Mixed meal: protein (20–40 g), carbs (0.5–1 g/kg), low fat	Amino acid availability during training; glycogen top-up; gastric emptying complete
Pre-training (30–60 min before)	Small: 15–30 g carbs + 10–15 g protein (if needed)	Blood glucose maintenance; blunts cortisol response
Intra-training (>90 min sessions)	30–60 g carbs/hour + electrolytes in water	Blood glucose maintenance; performance preservation in long sessions
Post-training (0–2 hours)	Protein: 0.3–0.5 g/kg; Carbs: 0.5–1.0 g/kg if glycogen-depleted	MPS elevation; glycogen resynthesis; the "anabolic window" is wider than advertised (hours, not minutes)
Pre-sleep (30–60 min before)	30–50 g casein or equivalent (cottage cheese, Greek yogurt)	Sustained amino acid release through the nocturnal fast (4–6 h)
Between meals (general)	≥0.4 g/kg/meal spacing 3–5 hours apart	Leucine threshold attainment per feeding; 3–5 meals optimal for MPS

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